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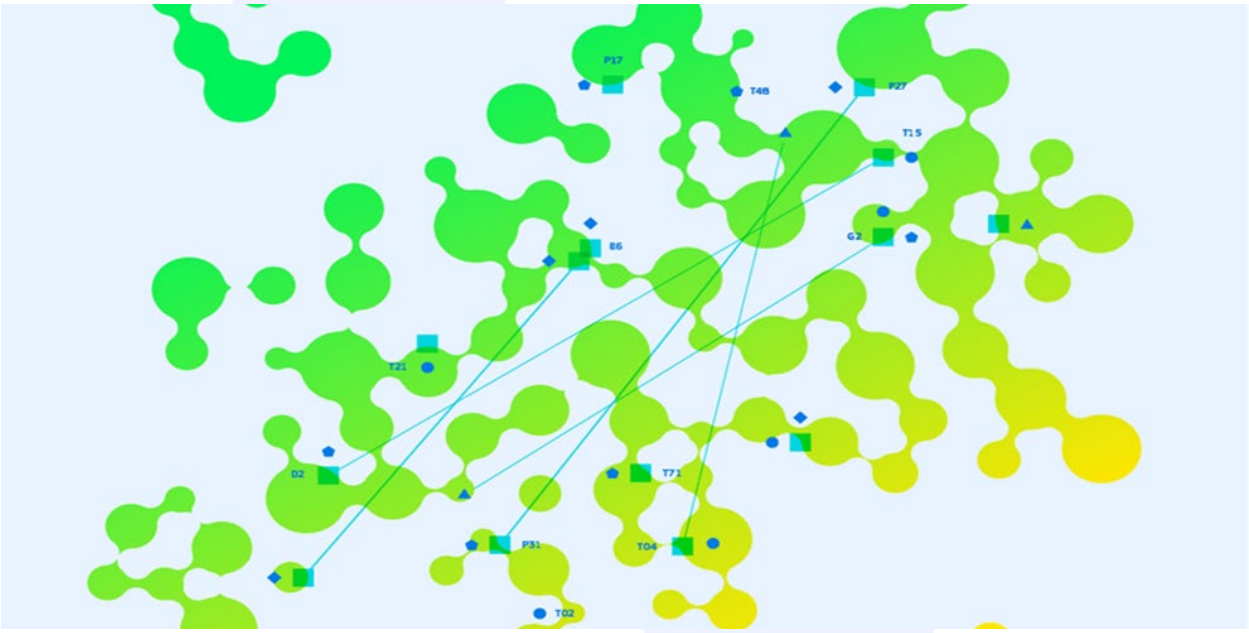
Joint Research Centre

Case Study of Andalusia space ecosystem: Unlocking regional potential through dual use and defence industry

Insights from the Exploratory Research Activity Regions Dual Use Open Strategic Autonomy (REGDUALOSA)

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Abstract

This report documents the findings of a review of the space ecosystem of Andalusia (Spain), carried out in 2025 within the framework of REGDUALOSA (Regions, Dual Use, Open Strategic Autonomy), an exploratory research initiative of the Joint Research Centre of the European Commission. The review was developed in partnership with the Regional Ministry of Industry, Energy and Mining and explores some of the policy pathways Andalusia might take to develop the space domain as a dual-use industry, thereby boosting regional competitiveness and contributing to European strategic autonomy. The analysis adapts the POINT methodology developed by the JRC, combining documentary research, expert interviews, and stakeholder consultations.

The study positions Andalusia within a volatile global landscape in which competitiveness, security, defence, and preparedness have emerged as central priorities of the European Union. Recent geostrategic tensions and the imperative of enhancing European defence capabilities, as underscored in the *White Paper on European Defence Readiness 2030* and the Readiness 2030 initiative, highlight the urgency of consolidating the European Defence Technological and Industrial Base (EDTIB). Andalusia, home to Spain's second-largest aerospace cluster, aspires to develop its space domain, while diversifying beyond its traditional reliance on aerostructures.

The report advances seven lines of action: scaling up the industry through defence integration and international expansion; streamlining access to finance; reinforcing industrial competitiveness; strengthening multi-level governance; fostering demand-driven innovation; and addressing talent shortages. Furthermore, it incorporates preparedness as a driver of transformative innovation and explores its interlinkages with dual-use technologies, promoting policy experimentation to address responses to climate-related disasters and hybrid threats.

The report thus provides a timely foundation for Andalusia's forthcoming steps. The participation of the Regional Government of Andalusia in the *Experimentation Journey on Territorial Preparedness*, under the Preparatory Action *Innovation for Place-Based Transformation* led by the Joint Research Centre and financed by the European Parliament, constitutes a unique opportunity to translate the insights of this report into new initiatives in cooperation with other European territories.

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Executive summary

In today's increasingly volatile global landscape, the European Commission has placed competitiveness, security and defence at the heart of its political agenda, introducing a renewed emphasis on the concept of preparedness—defined as the capacity to anticipate, absorb, and respond to complex crises. In this context, traditional border regions with a longstanding defence industrial base have a unique opportunity to strengthen their industrial fabric by addressing the EU's urgent need to enhance strategic autonomy, particularly in defence.

This report analyses a regional case study of Andalusia, a southern Spanish region with a deep-rooted tradition in the aerospace industry. It examines the Andalusian space ecosystem and explores how industrial and innovation policies are leveraging space technologies as a dual-use and defence-relevant domain, with the aim of positioning these technologies as a driver of regional competitiveness. The Andalusia case study was carried out by the Joint Research Centre in collaboration with the Andalusian Regional Ministry of Industry Energy and Mining.

The report was developed under the REGDUALOSA (Regions, Dual-Use, and Open Strategic Autonomy) Exploratory Research Activity, led by the European Commission's Joint Research Centre (JRC) between March 2024 and March 2025. REGDUALOSA examined how regional innovation and development policies can better support Europe's defence and dual-use industries. It is framed within the EU priorities for 2025–2029—Sustainable Prosperity and Competitiveness and A New Era of European Defence and Security—and aligns with the strategic vision set out in the Letta, Draghi, and Niinistö reports, and the Competitiveness Compass. The project focused on three case studies: Estonia (cybersecurity), Andalusia (space), and Podkarpackie (aerial robotics). These cases highlight different geographical sensitivities (the north-eastern border with Russia and Ukraine, and the southern European border with Africa), distinct dual-use technological domains, and varying levels of innovation (emerging, moderate, and strong) as well as development (less developed and transition regions).

Recent geostrategic tensions have further underscored the imperative to reinforce the EU's strategic autonomy and bolster its preparedness for emerging security challenges. As highlighted in the European Commission's White Paper on European Defence Preparedness 2030, the Union must build the capabilities necessary to ensure its own security and prosperity, supported by a resilient and innovative European Defence Technological and Industrial Base (EDTIB). The ReArm Europe-Readiness 2030 initiative, announced by the Commission, proposes mobilising up to €800 billion over four years to support increased defence spending among Member States.

Complementing these initiatives, the EU Defence Readiness Roadmap 2030—jointly published by the European Commission and the High Representative on 16 October 2025—sets out a comprehensive framework to strengthen Europe's defence capabilities and industrial readiness. Its Space for Defence dimension, which includes flagship actions such as the European Space Shield, explicitly recognises space as a strategic enabler of European security. The roadmap thus provides an essential policy reference for aligning regional dual-use and space-industrial strategies with the Union's broader defence preparedness agenda.

Andalusia, with over eight million inhabitants and a strong aerospace sector, represents, after Madrid, Spain's second-largest aerospace cluster. Although the space industry remains small in

relative terms (7.3% of the regional aerospace turnover in 2023), it has significant growth potential. The regional ambition, set out in the Aerospace Industrial Strategy 2027, is to capture in the midterm the 25% of the national space sector while diversifying beyond traditional aerostructures and fostering systemic spill-overs into the regional economy.

The analysis applies an adapted version of the POINT methodology (Projecting Opportunities for Industrial Transition), assessing four critical subsystems: Planning, Production, Knowledge, and Users. The research employs a mixed-methods approach, incorporating a thorough literature review and eighteen detailed expert interviews and a stakeholders' consultation session organised by the Andalusia government on 24 January 2025.

Andalusia's ecosystem combines SMEs, knowledge centres, and major primes such as Airbus, Indra and GMV. Strengths include satellite subsystems, ground segments, precision sensors, and software. However, the absence of an integrator in the region, limited private financing, and high barriers to entry constrain growth.
Funding relies heavily on public funding instruments (ESA programmes, Spanish Space Agency, Centre for Technological Development and Innovation-CDTI, Space Technological Plan-Recovery Resilience Facility), complemented by regional initiatives such as the incentive programme for participating in the life cycle of Satellites systems (launched by the Regional Ministry of Industry and Energy) and the Precommercial Public Procurement process, promoted by the Regional Ministry of University, research and Innovation, SPACE Innova, all them funded by European Regional Development Fund, the core resource to support regional innovation. Andalusia's space industry is characterised by advanced research infrastructure (Astrophysics Institute of Andalusia, Aerospace Vehicle and Atmospheric Research Testing Centre (El Arenosillo), Advanced Center for Aerospace Technologies CATEC, Seville Institute of Microelectronics, IMSE-CNM) Human capital development is progressing through new curricula (e.g. Master's Degree in Operation of Space Systems, micro-credentials in microelectronics in University of Seville and Malaga).

The establishment of the Spanish Space Agency headquarters in Seville in 2021, operational since 2023, serves as a strategic anchor for Andalusia's space ecosystem. Enhancing this positioning, the recent opening of the European Space Agency's Business Incubator Centre provides vital infrastructure to support space entrepreneurship and technology commercialization. The first call for applications by BIC ESA Andalusia was initiated in August 2025.

Interviews and stakeholders' consultation highlighted four key areas of improvement:

- Internationalisation: increased presence in ESA, EU and NATO programmes, and global markets.
- Financing mechanisms: more agile, less bureaucratic, and better adapted to SMEs.
- Talent development: specialised curricula, vocational training, lifelong learning.
- Defence integration: systematic participation in the European Defence Fund (EDF), European Defence Industry Programme (EDIP) and Digital Europe DIANA, and national defence programmes.

Additionally, a reality we cannot overlook is Europe's range of systemic challenges—such as climate change impacts, cyber-attacks and energy disruptions—alongside the conflict on European soil, which necessitates anticipation and proactive strategies. The development and implementation of space technologies can effectively address these issues, with applications in disaster management,

communication, and environmental monitoring. These technologies provide real-time data that is crucial for informed response and recovery, supporting governance models oriented towards foresight and innovation. Space technologies offer essential frameworks and opportunities for adaptation, allowing crises to be transformed into catalysts for progress. They facilitate the development of advanced systems, products, and services, thereby enhancing resilience and ensuring solutions that benefit human and environmental well-being.

The European Preparedness Strategy launched in April 2025 integrates anticipatory approaches into policy. For Andalusia, preparedness offers an opportunity to align space-based capabilities with territorial resilience needs. This includes wildfire prevention, flood and drought management, cyber-resilience and hybrid conflict response. Preparedness thus becomes a transformative innovation driver bridging civil, defence and environmental agendas. creating stronger links between territorial preparedness and territorial competitiveness.

The report outlines seven strategic lines of action designed to forge stronger connections between the Andalusian space ecosystem, territorial preparedness, and regional competitiveness:

1. **Scale up the space industry through international expansion and defence integration.** Reinforce participation in civil programmes within the ESA and European Commission's space initiatives by collaborating with the Spanish Space Agency (AAE). Explore opportunities in the defence market at national, European, and NATO levels. Facilitate SME participation in European value chains, promote a dual-use culture, and support cross-border cooperation.
2. **Streamline access to funding and incentives** by designing effective incentive programmes, reforming financing rules, and reducing administrative burdens. Leverage the STEP platform to create synergies between the European Regional Development Fund (ERDF) and other European programmes, such as Horizon Europe, European Defence Fund, the European Defence Industry Programme, and the Digital Europe Programme. Additionally, strengthen complementarities between the ESA, EUSPA, and national and regional programmes.
3. **Strengthen industrial competitiveness**, by scaling up regional companies, empowering clusters tied to strategic technologies, and reinforcing regional capacities in artificial intelligence and quantum technologies
4. **Enhance multi-level governance and whole-of-government coordination.** Stablishing a coordination board comprising members from AEE-ESA, INTA, and the Andalusian Regional Government while leveraging Andalusia's representation at national, European, and international levels. At the regional level, create a Space-Defence Coordination Task Force within the framework of the Andalusian Aerospace Strategy,
5. **Foster demand-driven innovation through public procurement** by utilising challenge-driven frameworks and providing capacity building for public buyers. Move beyond lowest-price criteria to prioritise performance, innovation, security, and strategic value. Encourage multilevel and cross-border collaboration, as well as the exchange of best practices in pre-commercial procurement (PCP) and pilot schemes focused on space.
6. **Address talent and skills** gaps by expanding higher education and vocational training. Strengthen space-specific curricula at Andalusian universities, encompassing degrees, master's programmes, and micro-credentials, with a focus on dual-use technologies. Establish lifelong training programmes for technicians and engineers in emerging fields such as AI, quantum technology, cybersecurity, and robotics. Develop joint training programmes that involve companies, universities, research centres, and the armed forces to bridge the knowledge transfer gap. Foster collaboration between Andalusian academia and the European Security and Defence College.

7. **Boost preparedness as a driver of transformative innovation** by developing policy experimentation processes to explore mission-oriented initiatives, such as combatting wildfires, managing water resources, and enhancing cyber resilience and addressing hybrid threats, in relation to dual-use and defence industries.

This report provides a timely foundation for Andalusia's next steps at EU level. The Regional Government's forthcoming collaboration in the Experimentation Journey on Territorial Preparedness, under the Preparatory Action "Innovation for Place-Based Transformation" led by the Joint Research Centre and financed by the European Parliament, represents a unique opportunity to translate the insights of this report into concrete pilot action in collaboration with other European Regions.

Introduction

In the face of growing global instability, the European Commission has made competitiveness central to its political agenda, focusing anew on preparedness—defined as the ability to anticipate, absorb, and respond to complex crises (Niinistö, 2024). Within this setting, traditional border regions (Kattel & Soete, 2024; Trippl et al., 2024) with established defence industries have a distinct chance to reinforce their industrial base by contributing to the EU's pressing goal of bolstering strategic autonomy, especially in the field of defence ((Kattel & Soete, 2024; Trippl et al., 2024)

Ongoing geostrategic tensions have further highlighted the necessity of strengthening the EU's strategic autonomy and improving its preparedness for new security threats. challenges (Ertel & Göler, 2025; Scazzieri, 2025; Schaible et al., 2025; Fiott, 2025). The European Commission's White Paper on European Defence Readiness 2030 (European Commission, 2025a) emphasizes that the Union must develop the capabilities needed to secure its prosperity, relying on a robust and innovative European Defence Technological and Industrial Base (EDTIB). The European-Readiness 2030[1] initiative, unveiled by the Commission, aims to mobilise up to €800 billion over four years to promote higher defence expenditures among Member States, amounting to approximately 3.5% of the EU's total GDP in that timeframe (Castilla, 2025)

The EU Defence Readiness Roadmap 2030[2]—jointly released by the European Commission and the High Representative on 16 October 2025—sets out a coordinated framework to strengthen Europe's defence capabilities and industrial preparedness. Its Space for Defence strand, including initiatives such as the European Space Shield, highlights the strategic role of space systems in underpinning European security and autonomy.

The communication EU communication "EU Defence Industry Transformation Roadmap: Unleashing Disruptive Innovation for Defence Readiness" [3]bring to the policy arena the learnings to the Ukraine experience in the battlefield" underscoring the centrality of agility, responsiveness, dual-use innovation and speedy integration of civilian technologies, highlighting the importance of nurturing agile ecosystems able to quickly provide operational solutions through distributed innovation networks like start-ups, civilian software communities.

Andalusia, a southern Spanish region and one of the country's primary aerospace hubs, is well-positioned to capitalise on this momentum. As a gateway between Europe and Africa, Andalusia benefits from a strategic location, a rich legacy in naval and aviation defence, and a growing commitment to diversify into the space sector. The region's distinctive assets—such as space infrastructure, government backing, and a proactive industrial base—present a compelling case for leveraging space technologies to enhance regional competitiveness, while contributing to Spanish national and Europe's strategic objectives.

[1] White paper for European defence - Readiness 2030 - Defence Industry and Space

[2] Readiness Roadmap 2030 - European Commission

[3] EU Defence Industry Transformation Roadmap: Unleashing Disruptive Innovation for Defence Readiness

The Andalusian's ambition, stated in its Aerospace Industrial Strategy 2027, is the aerospace sector being the driver of a new regional productive model towards competitiveness and sustainability, in which knowledge and innovation form the basis, with companies that make efficient use of territorial capital, are integrated into global value chains, and act responsibly towards Andalusian society and its environment. The objective of the space-related measures included in this programme is to ensure that the Andalusian space sector, which currently represents around 6%[4] of the national total, can expand to reach 25% in the medium term. To achieve this, the strategy seeks to increase the economic activity of the space industry in Andalusia, raise the number of regional companies engaged in space-related activities—thereby generating more direct employment—and strengthen the international projection of the sector by boosting the participation of Andalusian firms in European and global space projects[5].

The REGDUALOSA project—Regions, Dual-Use, and Open Strategic Autonomy—is an exploratory research initiative conducted by the European Commission's Joint Research Centre (JRC) between March 2024 and March 2025. Its primary objective was to explore how regional innovation and development policies can better support Europe's defence and dual-use industries within the broader framework of Open Strategic Autonomy.

Aligned with the strategic visions articulated in the Letta, Draghi, and Niinistö reports, REGDUALOSA contributes to evidence-based policy formulation in support of the Commission's 2025–2029 priorities: "Europe's Sustainable Prosperity and Competitiveness" and "A New Era of European Defence and Security."

The guiding research question was: How can defence and dual-use industrial investment be leveraged by territorial innovation policies to promote regional development within the EU's Open Strategic Autonomy framework?

To address this, the project developed three territorial case studies—Estonia, Podkarpackie (Poland), and Andalusia (Spain)—each focused on a distinct socio-technical system: cyber-defence, aerial robotics, and space technologies, respectively. The Andalusian case centres on space, a domain in which the region is building on its aerospace legacy to develop new strategic capabilities.

Why Focus on Space? Space technologies have become vital to the EU's economic resilience, geopolitical standing, and digital sovereignty. They support both civil and defence applications, contributing to secure communications, critical infrastructure protection, disaster management, border surveillance, environmental monitoring, and sustainable development. The EU's flagship programmes—EGNOS, Galileo and Copernicus—exemplify how space assets underpin preparedness, green and digital transitions, and global influence. The strategic dual-use value of space is further reinforced by initiatives such as IRIS² (Infrastructure for Resilience, Interconnectivity and Security via Satellite), aimed at enhancing secure EU-wide communications. This reflects a growing acknowledgement that space infrastructure is foundational to the EU's defence and civil capabilities alike. However, the Draghi report (2024) warns of Europe's increasing reliance on foreign nations

[4] https://andaluciaaerospace.com/wp-content/uploads/2025/07/Informe-del-sector-aeroespacial-andaluz-2024-1.pdf
[5] Estrategia Aeroespacial de Andalucía - Junta de Andalucía

for space access and high-end components, advocating for boosted domestic production and expanded industrial capacities in defence and space sectors.

The space sector has demonstrated remarkable dynamism in recent years, spurred by ambitious programmes like Artemis and the Lunar Gateway[6], which are pivotal initiatives aimed at advancing lunar exploration and fostering international collaboration. These programmes highlight the growing momentum and innovative potential within the sector, reflecting a global commitment to pushing the boundaries of space exploration and development.

According to the ESA Space Economy Report (2024) global institutional space budgets reached a record €106 billion in 2023—an 11% increase largely driven by defence spending. Although civil budgets still dominate in Europe (85%), defence allocations have overtaken civil spending globally for the first time. The United States remains the global leader, with China and Russia also significantly expanding their space capabilities. This shifting balance has prompted the EU to reaffirm its commitment to space as a core element of strategic autonomy.

A recent World Economic Forum and McKinsey & Company's report (2024) estimates that lower costs and improved access to space-enabled technologies such as communications; positioning, navigation and timing; and Earth observation services could take the global space economy to $1.8 trillion by 2035, up from $630 billion in 2023. According to EU Space Economy in figures (OECD, 2023) in OECD countries, space-based systems already support more than half of the most frequently designated critical infrastructures and services, such as transportation, energy, food supply and law enforcement. Space-based observations provide more than half of the essential climate variables that are used to monitor climate change, including for methane super-emitter events.

Beyond all these developments, space investment also fuels regional innovation and economic development. It enables technological spillovers across sectors and fosters high-value job creation, making it a cornerstone of future-oriented industrial policy.

Why Andalusia? Andalusia is Spain's second-largest aerospace hub after Madrid, with a turnover in 2024 of 2.914 Mil € and 14.496 professionals. The space and defence sectors are emerging areas of strategic focus within the region. This development is underpinned by a strategy of smart diversification, building on Andalusia's established aerospace tradition, robust industrial, academic and defence capabilities, and the active engagement of local stakeholders in identifying new specialisation pathways. It also reflects a timely response to favourable conditions at both national and European levels.

While Andalusia remains one of Spain's less economically developed regions and is classified as a moderate innovator in the European Innovation Scoreboard, it presents a distinctive combination of strategic location, institutional support, and industrial capacity. These factors make it an ideal candidate for a regional case study, offering a useful reference for other territories seeking to pursue new avenues for industrial development.

[6] Artemis - NASA

- Strategic Location: Situated at the EU's southern border, Andalusia adds a Mediterranean and south-border-region perspective to the overall territorial analysis, complementing the central and northern European cases.
- Institutional commitment. The Andalusian Regional Government has actively supported the aerospace and defence ecosystem over recent years through the Andalusian Aerospace Strategy 2027. This support includes, among other initiatives, infrastructure development, calls for projects targeting the satellite manufacturing value chain, and public procurement initiatives aimed at fostering innovation in the use of space technologies within the public sector
    - The Spanish government has established the headquarters of the Spanish Space Agency in Seville, further reinforcing the national commitment to space development in the region.
    - The Regional Ministry of Industry and Energy has served as the counterpart to the JRC in this study, providing consistent support throughout
- Aerospace and Defence Industrial Base: The region hosts Airbus Defence, NAVANTIA, INDRA, GDSB, EM&S, GMV, benefits from a strong network of SMEs and start-ups, and a leading role of the Andalusia Aerospace Cluster.
- Smart Specialisation: The regional Smart Specialisation Strategy (S4 Andalusia)[7] and the Aerospace Industrial Strategy 2027[8] explicitly prioritise space and dual-use technologies.
- Cross-Sector Collaboration: There is a demonstrated culture of collaboration among civil, defence, academic, and industrial actors, facilitating integrated innovation. The collaboration between the Andalusian Government and the Ministry of Defence in the strategic project to establish the Army's Logistics Base in Cordoba[9]- set to become an international benchmark in smart industry, innovation, and digital advancement, exemplifies the regional government's strong commitment to supporting the defence industry. It also serves as a model of inter-administrative and multi-level cooperation.

In sum, Andalusia exemplifies how regional innovation ecosystems can align with Spanish national and EU strategic priorities. By fostering space technologies with both civil and defence applications, the region stands to reinforce the European Union's preparedness and competitiveness while driving its own socio-economic development.

This document is structured into four main chapters. Following the presentation of the methodology, Chapter 1 delineates the scope and boundaries of the system under consideration. Chapter 2 analyses the current Andalusian space system, focusing on its actors, mechanisms, and framework conditions. Chapter 3 outlines the envisaged future state of the system, while Chapter 4 advances a set of recommendations and next stepts aimed at achieving this desired situation.

7 S4-Andalucía – Transformación económica Sostenible e Inteligente

8 Estrategia Aeroespacial de Andalucía. - Planes y programas - Junta de Andalucía

9 Junta, Ayuntamiento, Diputación y empresarios colaboran en el proyecto singular de formación de la Base Logística del Ejército en Córdoba - Noticias - Junta de Andalucía

Methodology

This case study report applies an adapted version of the Projecting Opportunities for Industrial Transitions (POINT) methodology (Pontikakis et al., 2020). It provides a systematic framework for framing, executing, and analysing industrial themes, focusing specifically on the space technology industry in Andalusia.

Utilising the POINT methodology, we have mapped four functional sub-systems of the current Andalusian space ecosystem:

- Orientation and planning
- Resource mobilisation
- Production of knowledge, goods, and services
- Consumers and users

Following this mapping, the study identifies the system components that are missing to achieve Andalusia's the desirable scenario. Subsequently, the report pinpoints leverage points and formulates policy-oriented recommendations.

The research employed a mixed-methods approach, incorporating a thorough literature review and eighteen detailed expert interviews. These interviews span industry experts from different-sized companies, governance officials (regional and national), universities, a technological centre, and key end users, including representatives from the Andalusian public administration and private users. Refer to Annex 1 for the full list of interviewees and questionnaires.

The preparatory works and literature review started in June 2024. The interviews were carried out between October 2024 and January 2025. To further validate and enrich the findings, a dedicated stakeholder consultation workshop was held on 24 January 2025 organised by the Regional Ministry of Industry, Energy and Mining. attended by over 40 experts (see agenda in the Annex 2). This session yielded critical insights into both the current state of the ecosystem and its potential future development from the perspective of regional actors.

Subsequently, the emerging results were shared with independent regional, national and international experts, who contributed through their reviews and comments, leading to the development of this final report.

1. Scope and boundaries of the system covered by the report

1.1. Andalusia region

Andalusia spans an area of 87,268 km^2**,** accounting for approximately 17.3% of Spain's total landmass. Geographically, the region extends from the Mediterranean Sea in the east to the Atlantic Ocean and the Portuguese border in the west, occupying a strategic position at the crossroads of Europe and Africa.

Figure 1 Andalusia (Spain)



Source: Adaptation of Encyclopaedia Britannica

With a population exceeding eight million**,** Andalusia faces enduring structural challenges, particularly in the labour market. In the second quarter of 2025, the region registered an unemployment rate of 14,85%[10]**,** among the highest in the European Union. Its GDP per capita (in purchasing power standards) in 2023 stood at €23.218[11] approximately 40% below the EU average, so from the point of view of European Cohesion policy, Andalusia is a less developed region.

According to the Regional Innovation Scoreboard (2025)[12,] Andalusia is classified as a moderate innovator, with a performance score equivalent to 76.4% of the EU average. In 2023, R&D expenditure amounted to 1.14% of regional GDP, and the private sector contributed 34.56% of total R&D investment.

Andalusia ranks second in Spain in terms of aerospace and defence turnover and employment**,** with major players such as Airbus Defence and Space, Navantia, General Dynamics Santa Barbara

[10] EPA de Andalucía 2025 | Datosmacro.com
[11] INEbase / Economía /Cuentas económicas /Contabilidad regional de España / Últimos datos
[12] ec_rtd_ris-regional-profile-es.pdf

(GDSB) playing a pivotal role in the regional industrial base. The region has also shown dynamism in high- and medium-technology exports**,** which reached a record €7.32 billion in 2023—representing a 21.6% year-on-year increase**,** outpacing the national growth rate[13].

Andalusia's location, abundant solar and wind energy, and access to critical minerals give it a natural edge in the net-zero transition (Tapia & Venturi, 2025).The region aspires to become a hub for sustainable industry and innovation.

1.2. Scope and boundaries of the system

The report examines dual-use space research, innovation, and industrial activity within the Andalusian aerospace Ecosystem, based on the scope defined by the Andalusian Space Sector Map[14] . This encompasses activities related to the development of launcher technologies, satellites, ground segments, satellite operators, and space sciences. In addition, it includes a wide range of application-oriented activities that exploit space data generated through Earth observation, navigation, and positioning programmes. It also identifies enabling technologies such as software, artificial intelligence, and advanced manufacturing. Nevertheless, given that space and defence industrial policy falls under national competence, together with the highly relevant European and multilateral dimensions (NATO and the European Space Agency) and the fact that the value chain extends well beyond regional borders, the analysis necessarily expands beyond Andalusia.

Following the Andalusian space sector map 2022, the total number of entities in the region with experience in the space sector reached 196. In terms of geographical distribution, the province of Seville hosts the largest share, with 42.86% of all entities, followed by Granada (15.82%) and Málaga (10.71%). University research groups account for 41.84% of the total, followed by private companies, which represent 36.22%.

Around 80 research groups are involved in space projects across the region, with notable contributions from the Universities of Seville, Málaga, and Granada, as well as the Institute of Astrophysics of Andalusia which manages the Sierra Nevada Astronomical Observatory, one of Spain's most significant space infrastructures.

While companies form a significant proportion of the ecosystem numerically, their levels of engagement and specialisation within the space sector may vary considerably. Among the most prominent companies in the Andalusian space sector are Alter Technology, Solar MEMS, DHV Technologies, Safran, and Integrasys. While not directly analysed in this report, a considerable number of innovative Information and Communication Technologies (ICT) companies—particularly those located in the technology parks of Málaga, Seville, and Granada—also play an important role for Andalusia, given their capacity to develop applications in the downstream segment of space services.

Andalusia also hosts testing facilities capable of supporting space activities, such as CEDEA-CEUS in El Arenosillo (Huelva) and ATLAS in Villacarrillo (Jaen). The latter is already being used for the launch of High Altitude Pseudo Satellites (HAPS) platforms.

[13] https://www.andaluciatrade.es/observatorio-internacionalizacion

14 Mapa Espacio Andalucia 2022_v1-FINAL con anexos.pdf

Preliminary data from Andalucía Aerospace shows that the Andalusian space industry reported a turnover of approximately €86.7 million in 2024, which is 24% higher than the previous year. Employment also rose by 19%, reaching 400 professionals. In 2023, Andalusian space accounted for around 8% of Spain's total space sector revenues. Within the regional aerospace economy, the space segment represented 7.3% of overall activity.

1.3. Andalusian's ambition for the space and defence ecosystem

Andalusia's ambition, articulated in its Aerospace Industrial Strategy 202715, positions the aerospace sector as a transformative driver of a new regional productive model grounded in competitiveness, sustainability, and innovation. This vision rests on leveraging knowledge and technology as foundational assets, fostering companies that make efficient use of territorial capital, integrate seamlessly into global value chains, and act responsibly towards Andalusian society and its environment.

Within the framework of the Aerospace Strategy, the space sector is recognised as a strategic growth domain for Andalusia, offering diversification beyond the traditional aerostructures industry16. The objective set in 2021 was to ensure that the Andalusian space sector—at the time representing 5% of the national space industry— expand to 25% in the medium term. Achieving this ambition requires not only scaling up the economic activity of the regional space industry and increasing the number of firms engaged in space-related activities but also consolidating direct high-skilled employment and reinforcing Andalusia's international projection through active participation in European and global space programmes.

The strategy also recognises the centrality of defence within the evolving geopolitical and security context. Anticipating these dynamics in 2021, Andalusia set the objective of expanding the participation of regional companies in European defence initiatives. This foresight is now confirmed by the heightened strategic importance of defence at both at both national and European levels, reinforcing Andalusia's positioning at the intersection of space, aerospace, and dual-use innovation.

1.4. Reasons for boosting dual use space ecosystem as a competitiveness driver

Diversification and added value. Andalusia is Spain's second-largest aerospace hub, but 50% of its turnover is concentrated in low value-added activities (aviation aerostructures) and on one main contractor, AIRBUS defence and Space.

In 2024 the Andalusian aerospace sector comprises 148 companies, generating a turnover of €2.914 billion and employing 14,496 professionals with highly levels of qualification. The industry is primarily concentrated in the province of Seville, which hosts 106 companies, followed by Cádiz with 23. In 2023, the sector contributed 1.32% to Andalusia's GDP, reflecting its growing economic significance and the 13,84% of the Andalusia's industry GDP. The main activity is related the aeronautic industry (77%), mostly Airbus civil and Defence programmes (95%). 68% of the Andalusian aeronautic turnover comes from the Defence market. Andalucía Aerospace (2024)

[15] Estrategia Aeroespacial de Andalucía - Junta de Andalucía
[16]

reports that aerostructures represent 50% of the regional aerospace cluster, a comparatively low value-added segment within aeronautical manufacturing[17].

Nowadays, although small in size, the Andalusian space sector has a solid foundation and a clear international focus, cutting-edge research and testing infrastructure (INTA's CEDEA/CEUS, CATEC, IMSE-CNM) and is in a position to attract private capital, scale up its SMEs and consolidate its position as a competitive hub in the European space market.

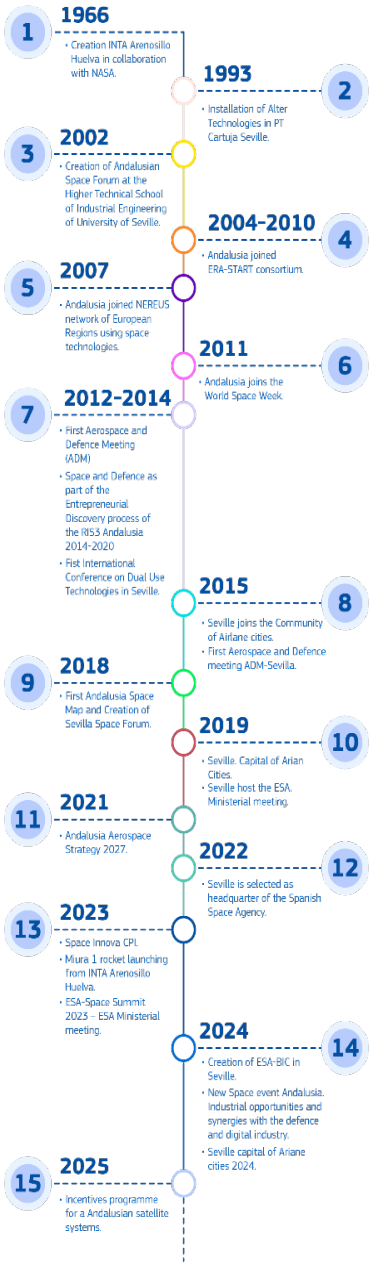
The history of the Andalusian space sector began in 1966, when NASA requested a site from the Spanish government on which to build a launch pad for meteorological rockets. The chosen location was an area called El Arenosillo on the coast of Huelva in eastern Andalusia, nowadays the location of the Spanish Institute of Aerospace Technologies (INTA)[18]. in 1993 took place the establishment of its first space company, Alter Technology, at the Cartuja Scientific and Technological Par (now SevillaTech). Around the same time, interest in space technologies was growing within Andalusian universities. This momentum led to the launch of the Andalusian Aerospace Forum in February 2002, initiated by the Higher Technical School of Industrial Engineering of University of Seville, with the support of institutions and individuals connected to the fields of aeronautics and space. Since then, the Andalusian space activity is growing, and its milestones are highlighted in figure 2. Over this period, several noteworthy developments have occurred: Seville hosted two ESA Ministerial Conferences in 2019 and 2023; it was selected as the capital of Ariane cities in 2019 and 2024; and the regional government has actively promoted and attracted space investment through events such as the Aerospace and Defence Meetings and the International Conference on Dual Use and Defence. Additionally, in 2021, the Andalusian Aerospace Strategy 2027 was launched and has since been implemented through various calls, procurements, and infrastructure investments. In 2023, the selection of Seville to host the headquarters of the Spanish Space Agency, and in 2024, Seville's successful bid to host the ESA BIC, marked important milestones in the history of Andalusia's space sector

Capitalising on a high-growth sector. The space sector is projected to grow ninefold by 2030, with a total market value expected to reach USD 1.8 trillion by 2035[19]. Europe currently commands 33% of the upstream and 19% of the downstream global space markets. However, private investment is disproportionately low, at €1.5 billion in 2024 (22% of global private spending). Andalusia could attract targeted private investment and scale-up regional SMEs, addressing this imbalance within the EU.

Space is a competitive industry and generates high-quality employment. The Spanish space industry , ranks fifth in Europe, following France, Germany, Italy, and the United Kingdom. In 2023, the sector achieved a turnover of €1200 million, contributing 1.46% to the country's industrial GDP. Spanish Space industry invests 14.4% of their turnover in R&D and its productivity is 1.88 times the equivalent to the general Spanish industry . Notably, 90% of Andalusian space products are exported to key international markets, including Europe, the United States, South Korea, Japan, and Latin American countries such as Chile and Colombia. In Andalusia, 44% of the employees in the aerospace sector are University graduates and 40 % vocational education and training graduates and qualified operators

[17] Informe-del-sector-aeroespacial-andaluz-2024-1.pdf
[18] https://www.inta.es/INTA/es/quienes-somos/historia/
[19] 'Space: The $1.8 Trillion Opportunity for Global Economic Growth', Insight Report World Economic Forum, April 2024.

Figure 2 Main milestones of Andalusia space Industry



Source: Own elaboration

Driving Regional Economic Development and Cohesion. Andalusia remains a moderate innovator and one of the least economically developed regions in Spain, with high unemployment and GDP per capita significantly below the EU average. Targeted investment in high-tech sectors like space and defence can support regional reindustrialisation, quality job creation, and reduce economic disparities.

Participation in EU strategic initiatives would also support Andalusia's Smart Specialisation Strategy for Sustainability (S4) and Aerospace Strategy 2027, which prioritise dual-use and advanced technologies. Leveraging the space sector would help align local priorities with EU funding and policy tools.

Aligning with the EU strategic priorities. Based on the key messages from EU Space Strategy for Security and Defence, the reports of Letta (2024) Draghi (2024), and Niinistö (2024), the Competitiveness Compass (European Commission, 2025b), the legislative proposal of the EU Space Act[20] and the Communication on the Space Economy (European Commission, 2025c), clearly positioned space technologies alongside AI, semiconductors, and clean energy as core to the "technologies of tomorrow.

The Draghi, Letta, and Niinistö reports explicitly identify space as essential to the Union's future economic security and resilience. By investing in dual-use and space capabilities, Andalusia contributes to reducing European dependencies and strengthening preparedness.

Geopolitical window of opportunity in the space and defence market. NATO member states have committed to reaching 5% of GDP on both core defence requirements (3,5 %) and defence and security-related spending (additional up to 1,5 %) by 2035, as agreed in The Hague in June 2025.

In conclusion, Andalusia has both the strategic rationale and the institutional foundations to become a key regional player in Europe's evolving space and defence landscape.

[20] https://defence-industry-space.ec.europa.eu/document/download/0adeee10-af7a-4ac1-aa47-6a5e90cbe288_en?filename=Proposal-for-a-Regulation.pdf

2. The current system: Actors, mechanisms, and framework conditions

In the analysis of the Andalusian space system, following POINT methodology, four main subsystems are identified:

- Orientation and Planning
- Resource Mobilisation
- Production
- Users, consumers.

2.1. Orientation and planning subsystem

In strategic terms, the evolution of the global space industry is largely determined by the priorities and resources of the principal space powers. In the United States, strategic direction is set by the White House, the Department of Defence and NASA. Their decisions are operationalised through extensive budgetary allocations, amounting in 2023 to approximately USD 73 billion—almost five times the investment of the European Union, which stood at USD 15 billion. These resources underpin programmes that exert a decisive influence on the global sector (European Commission, 2025c). Major US corporations such as SpaceX, Amazon Kuiper and Lockheed Martin further reinforce this influence by shaping technological standards and market dynamics. In the case of China, planning is highly centralised within state structures, aligning space policy with military, scientific and economic objectives

The international governance of standards remains strongly conditioned by fora in which the United States and China exert significant influence. By contrast, Europe's role is distributed between the European Commission, the European Space Agency (ESA) and the Member States.

NATO, though not an actor in industrial policy, plays a pivotal role in space security by ensuring interoperability and security standards across its members, thereby indirectly influencing the technical frameworks subsequently adopted by national and European industries.

On the civil side, Spain participates in ESA through the recently established Spanish Space Agency (AEE), headquartered in Seville[21]. This development represents a significant strategic asset for Andalusia, further enhanced by the establishment of the ESA Business Incubation Centre (BIC-ESA) in the region. At the national level, the elaboration of a Spanish Space Strategy and the drafting of a Spanish Space Law are currently underway.

Within Spain's decentralised political system, Andalusia possesses substantive competences in industrial policy, innovation and regional development, as articulated in its Statute of Autonomy. The regional government's instrument for planning, executing and evaluating public actions to be carried out in the Aerospace sector in Andalusia is the Aerospace Strategy, approved by the regional council in 2021.

[21] In November 2025 there will be the ESA Ministerial Conference in Bremen (Germany), where funding to space programmes in ESA member countries will be decided

In order to understand the different levels of governance that have influence, direct or indirect, in the Andalusian space ecosystem, the following layers are being unfolded: outer EU, European Union, Nacional y Local., illustrated in Figure 3.

Figure 3. Orientation and Planning subsystem



Source: Authors based on the interviews to the experts

2.1.1. Orientation and planning actors

Outer EU

While the regulation of outer space originated in the 1967 under the auspices of the United Nations—establishing space as the common heritage of humankind and reserving its use for peaceful purposes—the EU did not acquire formal competences in space until the Treaty of Lisbon (2009). Articles 4(2) and 189 of the Treaty on the Functioning of the European Union (TFEU) mandate the EU to support and coordinate Member States' efforts in space, while explicitly prohibiting harmonisation. Prior to this, cooperation took shape through intergovernmental organisations, culminating in the founding of the European Space Agency (ESA) which includes members which are not part of the EU.

European Space Agency[22]

The European Space Agency (ESA) is an independent intergovernmental organisation founded in 1975 to promote peaceful cooperation in space science, technology, and applications among European countries. Its key objectives are to establish a unified European space policy, coordinate national and European programmes—progressively integrating national initiatives—develop and recommend industrial policy, and manage both mandatory and optional space activities. The ESA's members currently include 22 European states and associate members such as Slovakia, Latvia and Lithuania. Canada participates through specific cooperation agreements, while other states hold collaboration agreements. The European Space Agency (ESA) Council meeting at Ministerial level (CM25) will be held in Bremen, Germany on 26 and 27 November 2025 to determine Europe's funding priorities for the coming years.

ESA is distinct from the European Union (EU): although both aim to advance European interests in space, their legal frameworks, membership, competences, and governance differ. ESA and the EU cooperate closely, as formalised by the Financial Framework Partnership Agreement (FFPA)—last signed in 2021—which establishes the roles, funding, and collaboration mechanisms between both organisations.

ESA programmes are financed through mandatory contributions from members (based on GDP) and optional programmes to which members contribute voluntarily. As of 2025, ESA's budget is €7.68 billion. The agency's geographical return policy ensures member states receive contracts commensurate with their financial contributions, supporting national industries and technological competitiveness. Spain's rising stake—now €932 million—has bolstered its space industry's leadership and integration into ESA, EU, and global markets. Thanks to ESA's geographical return policy, Spanish investment is channelled into industrial contracts, boosting the country's technological capabilities and competitiveness in ESA programmes, and positioning its companies for success in EU, EUMETSAT, and global markets.

ESA also drives space standardisation and compliance through the European Cooperation for Space Standardisation (ECSS), shaping certification and protocols in European space missions.

[22] ESA - ESA facts

North Atlantic Treaty Organisation (NATO)

Spain is among the 23 European Union Member States that are also members of NATO.

NATO[23] recognised space as a strategic operational domain for collective defence in 2019, formally designating it as the Alliance's fifth domain—alongside land, sea, air, and cyber—and establishing the NATO Space Centre within Allied Air Command to coordinate space-related activities.

Through the Defence Planning Process (NDPP), NATO identifies the capabilities it requires and promotes their development by Allies, providing a framework to harmonise national defence plans with NATO's—without compromising national sovereignty. Although NATO defines space capabilities within the NDPP, it does not currently assign specific space capability targets to individual Allies.

NATO, develops standardisation agreements known as STANAGs (Standardization Agreements). These agreements are crucial for ensuring interoperability among the armed forces of NATO member countries. NATO's work on STANAGs encompasses various aspects of space operations, including technical specifications, operational procedures, and safety standards, thereby facilitating seamless cooperation in multinational missions.

In February 2025, NATO Defence Ministers endorsed the organisation's first Commercial Space Strategy, a landmark initiative designed to deepen cooperation with the commercial space sector. The strategy aims to enhance Alliance capabilities in areas such as space situational awareness, intelligence, surveillance, and reconnaissance, and to support interoperability through the creation of a Space Capabilities Group under the Conference of National Armaments Directors. It also introduces a "space front door" mechanism to facilitate structured engagement between NATO and industry.

NATO's industrial engagement is further reinforced through the NATO Innovation Fund, which invests in emerging technology companies relevant to defence and security. This investment activity complements the Defence Innovation Accelerator for the North Atlantic (DIANA)—operational since June 2023—which supports dual-use innovation by linking technology developers with a network of test centres and accelerators across the Alliance.

European Union Level

European Commission

The EU's investment in space infrastructure began in 1999 with Galileo, its global navigation satellite system, and Copernicus, providing Earth observation capabilities. Both offer free, open-access data to governments, businesses, and citizens. The EU Space Programme Regulation (Regulation (EU) 2021/696), adopted in 2021, allocated nearly €15 billion under the 2021–2027 Multiannual Financial Framework (MFF). In the same year, the European Defence Fund was created to incentivise and support collaborative, cross-border research and development in the area of defence, allocating € 8 billion of the MFF. The Directorate-General for Defence Industry and Space (DEFIS) and the European Union Agency for the Space Programme (EUSPA) were established to coordinate policy and ensure secure management of EU space assets. New initiatives—such as

[23] https://www.nato.int/cps/en/natohq/topics_175419.htm

IRIS², Space Traffic Management (STM), and CASSINI (€1 billion for SMEs and start-ups)—demonstrate the EU's intent to strengthen competitiveness, security, and innovation.

At European level, amid growing geopolitical tensions, the Strategic Compass (2022) identified space as a strategic domain. Derived from the Strategic Compass, the EU Space Strategy for Security and Defence (2023) stressed that space infrastructure is critical to EU security and must be protected. IRIS²—the EU's third flagship space project—will deploy a multi-orbital constellation of around 290 LEO and MEO satellites to provide secure communications for government (defence, crisis management, border surveillance, diplomatic services) and commercial broadband in underserved areas. First services are planned for 2030.

In 2024, the European Commission adopted the European Defence Industrial Strategy (EDIS). On 16 October 2025, the Council and the European Parliament reached a provisional agreement about the associated European Defence Industry Programme (EDIP). This programme involves €1.5 billion, in the form of grants. The total budget includes €300 million in support of Ukraine's defence industry. Moreover, the Council and the Parliament invited the Commission to explore options for additional financial resources in the future.

For the 2024–2029 European Commission's mandate, the vision for the European space economy (European Commission, 2025a) and the EU Space Act are relevant topics. The Space Act[24] responds to concerns over fragmented regulations that hinder innovation, raise costs, and reduce Europe's market share. It proposes an EU-wide framework to ensure safety, resilience, and sustainability, enabling companies to scale across borders. Introduced in June 2025, the Act's three pillars—Safety, Resilience, and Sustainability—aim to cut red tape, protect space assets, and create a predictable business environment. The proposal is now under consideration by the European Parliament and Council via the ordinary legislative procedure.

In the current mandate of the European Commission, which for the first time counts on a Commissioner on Defence, the Joint White Paper-READINESS 2030 has defined the guidelines for the European Defence industrial policy.

European Union Space Programme Agency EUSPA[25]

EUSPA is responsible for the operational management of the EU Space Programme and offers cutting-edge positioning, navigation, and timing services through Galileo and EGNOS, along with satellite communications for GOVSATCOM and soon, IRIS². It also enhances the use of data from these systems, Copernicus, and upcoming IRIS² services across multiple domains. Furthermore, EUSPA fosters a dynamic European space ecosystem by providing market intelligence and technical expertise, leveraging Horizon Europe and other funding mechanisms. It ensures the security of the EU Space Programme's components both in space and on the ground through the Galileo Security Monitoring Centre (GSMC). The agency's Security Accreditation Board acts as the security accreditation authority for all components of the EU Space Programme

[24] European Commission adopted this legislative proposal for a Regulation on the safety, resilience, and sustainability of space activities in the Union on June 2025 which will be negotiated under the ordinary legislative procedure by the European Parliament and the Council.
[25] EU Agency for the Space Programme

European Defence Agency (EDA)

The European Defence Agency (EDA)[26], as part of the EU's Common Security and Defence Policy (CSDP), guides capability requirements and fosters research, technological development, and industrial autonomy in space. In charge of the coordinated Annual Review on Defence, or CARD, EDA supports Member States in their efforts to implement the EU agreed Capability Development Priorities by providing them an overview and analysis of the EU defence landscape and proposing recommendations for collaborative opportunities in capability development and acquisition., as much as possible synchronised with the NATO Defence Planning Process (NDPP). Additionally, EDA contributes to standardisation efforts by promoting collaborative projects among EU member states, focusing on the development of common standards for defence technologies. It acts as a bridge between Member States' ministries of defence and industry to generate collaborative research and innovation projects, in coordination with instruments such as Permanent Structured Cooperation (PESCO)[27] and the European Defence Fund (EDF).

EDA's Defence in Space Forum, launched in October 2022, accelerates implementation of the 2023 EU Capability Development Priorities in space—covering key areas such as space situational awareness, satellite communications, Earth observation, positioning, navigation and timing, access to space, and the protection of space systems

Moreover, EDA contributes to strengthening the European Defence Technological and Industrial Base (EDTIB) through research and development programmes that address technological dependencies in strategic sectors, including space. It works in close consultation with the European Space Agency (ESA) and the European Commission to identify priority technologies and support efforts toward autonomy and operational sovereignty.

European External Action Service (EEAS)[28]

The High Representative (HR) of the Union for Foreign Affairs and Security Policy manages the Common Foreign and Security Policy and Common Security and Defence Policy (CFSP/CSDP) aspects of the EU's space activities and ensures the consistency of the EU's external action in the space domain.

The High Representative exerts an operational role in response to possible threats to or through the systems and services deriving from the EU Space Programme and concerning the EU Satellite Centre. (SatCen)

In March 2023, the High Representative and the Commission issued the first EU Space Strategy for Security and Defence that will allow the EU to protect its space assets, defend its interests, deter hostile activities in space and strengthen its strategic posture and autonomy.

The EU Defence Readiness Roadmap 2030[29]—jointly released by the European Commission and the High Representative on 16 October 2025—sets out a coordinated framework to strengthen Europe's defence capabilities and industrial preparedness. Its Space for Defence strand, including initiatives such as the European Space Shield, highlights the strategic role of space systems in underpinning European security and autonomy. Together, these initiatives reflect a broader shift towards

[26] eda.europa.eu
[27] https://www.pesco.europa.eu/
[28] Space | EEAS
[29] Communication Preserving Peace-Defence Readiness Roadmap 2030

integrated civil-defence planning, positioning the space domain as both a driver of competitiveness and a key component of Europe's preparedness agenda.

Spain. National Level

Spanish Space Agency (Agencia Espacial Española, AEE)

AEE[30], created in 2023 in Seville, is a state body under the joint authority of the Ministry of Science and Innovation and the Ministry of Defence. It was established to integrate and coordinate Spain's space activities, providing a single national voice in European, bilateral, and multilateral fora, including the EU and the European Space Agency (ESA). Its remit includes the design and implementation of the National Space Strategy, the development of plans and programmes aligned with strategic objectives, the promotion of industrial, scientific, technological, and infrastructure capabilities, the preparation of a National Space Policy and a draft Space Act, and the mapping of Spain's space capacities. The Agency's core competences include the coordination of Spain's participation in the European Space Agency (ESA), the European Union's Space Programme, and other international initiatives.

The AEE is mandated to ensure the efficient allocation of public resources, monitor industrial returns, and promote technological capabilities, industrial competitiveness, and innovation across the national space sector. Its remit also extends to supporting the commercial exploitation of space assets, services, and data, thereby fostering entrepreneurship, start-up development, and the growth of space-related value chains.

The Agency manages major national space programmes, notably:

- Space Surveillance and Tracking (S3T): A service providing collision avoidance, re-entry prediction, and fragmentation analysis, implemented in cooperation with the European Space Agency and jointly managed by civil and defence authorities. The Spanish Space Surveillance and Tracking Surveillance Radar (S3TSR), It is a low Earth orbit (LEO) surveillance radar operating in the L band. The radar is installed at the Morón de la Frontera air base (Seville).
- Atlantic Constellation (ESCA) The project was initially a partnership between Portugal and Spain, to develop a constellation of 16 Earth observation satellites that would be used for ocean, Earth, and climate monitoring. United Kingdom committed to adding one additional satellite to the constellation.

It plays a pivotal role in the execution of the PERTE Aeroespacial, launched in March 2022 as part of Spain's Recovery, Transformation, and Resilience Plan (PRTR), is a high-impact, inter-ministerial initiative aimed at advancing Spain's aerospace and space capabilities across multiple strategic dimensions. Initially endowed with a public budget of €2.193 billion, the project's funding has since risen to approximately €2.5 billion through additional State contributions.

The AEE is tasked with ensuring that Spain's contributions to ESA and other international programmes are matched by proportional industrial returns, in line with the principle of geographical return.

The Spanish Space Agency (AEE) is currently designing the Spanish Space Strategy, which, in coordination with other competent bodies, will support actions to meet the space-related security

[30] Agencia Espacial Española

and defence objectives set out in the National Security Strategy (ESN), the EU Space Strategy for Security and Defence, and equivalent NATO documents

On 5th November 2025, the AEE has launched the preliminary public consultation on the draft Spanish law on Space Activities[31].

Ministry of Defence. Air and Space Force

The Ministry of Defence (MINISDEF) is the department of the General State Administration responsible for the preparation, development, and execution of defence policy. In September 2024 MINISDEF created the new General Direction of Strategy and Innovation[32] which is responsible for the planning and development of defence industrial policy and industrial cooperation in this area, international cooperation and representation in international defence industry forums, institutional support for the internationalisation of the Spanish defence industry and exports, control of foreign trade in armaments and equipment, technology transfers to third countries.

The renaming of the Spanish Air Force to the Air and Space Force (Real Decreto 524/2022) reflects the growing importance of outer space in national security and defence, highlighting the Armed Forces' essential role in protecting and controlling the space domain, with the Air and Space Force leading in this environment. To address the increasing vulnerability of the space domain, the Armed Forces have adapted their structure to ensure free access to, and use of, space, as well as the protection of national aerospace infrastructure, assets, and services. This adaptation included the creation of a Space Command within the Air and Space Force, tasked with centralising unit preparation and coordinating surveillance, control, and operational activities in space.

The Industrial and *Technological Plan for Security and Defence*, approved by the Council of Ministers in April 2025[33], constitutes Spain's response to the current international context. The plan includes the launch of 31 new Modernisation Programmes[34], which are added to the 56 already in progress. Among the explicitly space-related initiatives in the Modernisation Programmes are SEOT PAZ II, a programme launched in 2025 that will deploy two synthetic aperture radar (SAR) satellites for Earth observation, and the SpainSat NG I & II system, two geostationary satellites providing secure military communications in X, Ka and UHF bands. Together, these programmes represent Spain's main national investments in space capabilities, not just for defence, but national flagship programmes. In addition, several other modernisation programmes—such as Multidomain Cryptographic Capacity, the Integrated Advanced Cybersecurity System, and Secure Connectivity projects—are not dedicated space assets but are space-enabled, relying heavily on satellite communications and secure networks to achieve their objectives.

The *National Aerospace Security Strategy (ESAN 2025)*[35] —adopted by the National Security Council on 14 July 2025 and published through Ministerial Order PJC/845/2025 of 29 July—sets out the strategic vision, priorities, and coordination mechanisms for protecting Spain's aerospace environment. It recognises the need for coordination across all levels of government, explicitly highlighting the involvement of Autonomous Communities with competencies in areas such as

[31] https://www.aee.gob.es/Noticias/2025/noviembre/consulta-publica-anteproyecto-ley.html
[32] Disposición 18185 del BOE núm. 220 de 2024
[33] 230425-plan-industrial-y-tecnologico-para-la-seguridad-y-la-defensa.pdf
[34] Los 31 programas del plan de Defensa de Sánchez: entrenador avanzado, helicópteros, modernización de fragatas, obuses o vehículos anfibios
[35] Disposición 16214 del BOE núm. 187 de 2025

security, industry, and innovation. The strategy frames these regional authorities as essential partners in implementing national objectives, particularly in protecting aerospace assets, fostering industrial capabilities, and contributing to security-related R&D.

Several lines of action within the strategy—such as strengthening cybersecurity, securing aerospace infrastructure, and developing dual-use technological capabilities—require the integration of regional resources and expertise. This collaborative approach ensures that national aerospace security priorities are adapted to, and reinforced by, the strengths and specific contexts of each Autonomous Community, including Andalusia.

Ministry of Industry and Tourism

The Ministry of Industry and Tourism is the Department of the General State Administration responsible for proposing and implementing government policy on industry and tourism. This includes, among other aspects, industrial strategy and development, entrepreneurship, and small and medium-sized businesses. The Ministry of Industry and Tourism acts as a key financial stakeholder in the early stages of the development and manufacturing of major defence-related space systems.

Ministry of Science, Innovation and Universities

In Spain, responsibilities for science, innovation and universities are shared between the central government and the Autonomous Communities (with regional governments). The Ministry of Science, Innovation and Universities, in coordination with the Ministry of Defence, holds responsibility for the oversight and strategic direction of the Spanish Space Agency.

The Ministry of Science, Innovation and Universities sets national policy, coordinates research and university strategy, and represents Spain internationally. Meanwhile, the Autonomous Communities manage the organisation, funding, and operations of universities within their territories, approve university statutes, and regulate academic programmes—while respecting university autonomy.)

Regional Level. Andalusian Government

The governance of Andalusia's aerospace ecosystem operates under a dual-ministry framework defined by Decree 158/2022 of 9 August, which assigns to the Regional Ministry of University, Research and Innovation responsibilities for fostering research, technological innovation, knowledge transfer, and entrepreneurship. Complementarily, Decree 6/2024 of 29 July entrusts the Regional Ministry of Industry and Mining with the direct competences for the industrial promotion, regulation, and strategic development of the aerospace and defence industries. This institutional arrangement integrates innovation policy with industrial strategy, creating a coordinated approach to the advancement of the regional aerospace ecosystem.

Andalucía TRADE is the Regional Development Agency of the Andalusian Government. It is responsible for business development, innovation and knowledge transfer, as well as promoting Andalusian companies abroad and attracting investment and competitiveness. It reports to the Regional Ministry of Economy, Treasury and European Funds (Business Support and Internationalisation) and the Regional Ministry of the Presidency, Social Dialogue and Administrative Simplification. Andalucía TRADE has a long track record of dedication to the promotion and development of the aerospace sector.

A key focus is the internationalisation of the Andalusian economy, helping local businesses expand into global markets and supporting their export activities. Andalucía TRADE also provides financial support through subsidies and other instruments to facilitate business growth. Additionally, it promotes and energizes the business community through awareness campaigns, training programmes, and various initiatives designed to strengthen the regional entrepreneurial ecosystem.

Andalusia has a Regional Aerospace Industry Strategy 2021-2027[36] approved by the Regional Government's Council of Ministers in 2021, with responsibilities shared between the two mentioned regional ministries.

The Andalusia Aerospace Industrial Strategy was designed and is implemented through a participatory, interdepartmental approach led by a Steering Committee. This includes the General Secretary for Research and Innovation (Regional Ministry of University, Research and Innovation) and the General Secretary for Industry and Mining (Regional Ministry of Industry, Energy and Mining). Other regional ministries—such as the Regional Ministry of Economy, Treasury and European Funds (business support and internationalisation) and the Regional Ministry of the Presidency, Social Dialogue and Administrative Simplification—are represented by the Andalusian Regional Development Agency, TRADE, alongside representatives for regional ministries of education, vocational training, mobility, and transport. The business sector is represented by the Andalusian Confederation of Business, while workers are represented by the trade unions CCOO (Comisiones Obreras) and UGT (Unión General de Trabajadores).

The strategy includes a programme to finance the development of the regional space industry, with a particular focus on strengthening technological capacities aligned with the "New Space" paradigm. This encompasses R&D and industrial projects aimed at advancing cutting-edge technologies such as High-Altitude Pseudo-Satellites (HAPS), small satellites, microelectronics, radiofrequency, optronics, photonics, and quantum technologies. It also proposes measures for the integration of more Andalusian companies into new programmes of the European Space Agency (ESA).

In parallel, the Smart Specialisation Strategy for Sustainability of Andalusia S4 Andalucía (2021–2027) explicitly recognises space and dual-use and defence technologies as regional specialisation areas within the broader RIS3 Andalucía framework. The space sector is referenced under Priority 2: Consolidation of the advanced transport industry, with specific lines of action including, advanced manufacturing, research and innovation in new materials and development of innovative products for the transportation industries

Local Level. Seville and Malaga City Councils

While interest in new technologies spans the entire Andalusian ecosystem as shown in figure 9, it is in Seville and Málaga where the greatest concentration of industrial activity – number of firms- related to the space sector can be found)

Seville, The Administrative Capital of The Andalusia region, is a key hub for the regional space sector. It hosts 42.86% of all Andalusian entities with experience in space-related activities. The

[36] Estrategia Aeroespacial de Andalucía. - Planes y programas - Junta de Andalucía

first Andalusian space company, Alter Technology, was established in Seville Tech Park in 1996, and the city hosts now the headquarters of the Spanish Space Agency.

The City Council has actively positioned Seville as a prominent space city. In both 2019 and 2024, Seville held the presidency of the Community of Ariane Cities (CVA). In 2023, it formally submitted its bid to host the Spanish Space Agency. Following a competitive process, in 2025 Seville secured the establishment of a European Space Agency (ESA) business incubator for technology-based start-ups specialising in the space sector.

The city and its surroundings are home to major aerospace facilities, including the Aerópolis Aerospace Technology Park and Seville Tech Park, where most space-related companies in the area are located. The City Council is also developing a new urban planning initiative, the "Seville Aerospace District", aimed at further consolidating the city's role in the aerospace and space industries.

Seville has hosted key European space governance events, including the ESA Ministerial Council in 2019 and 2024, the latter during its CVA presidency. In November 2023, the city also hosted the ESA Ministerial Council opening European Space Week, followed by a joint ESA–EU meeting addressing space sustainability, the green transition, and autonomous access to space.

Málaga has emerged as an important node in Andalusia's space and advanced technology ecosystem, supported by its strong innovation infrastructure and municipal commitment to fostering high-tech industries. The city hosts the Málaga TechPark. Málaga also host the annual Small Satellites & Services International Forum (SSSIF), a major international meeting point for the small satellite.

The Málaga City Council,—such as the new IMEC semiconductor design and manufacturing centre, supported by the Ministry of Science, Innovation and Universities and the Andalusian Regional Government, plays an enabling role by supporting technology investment projects —and by facilitating the urban and administrative frameworks that strengthen the city's position as a competitive hub for space-related activities.

2.1.2. Expert Insights and SWOT

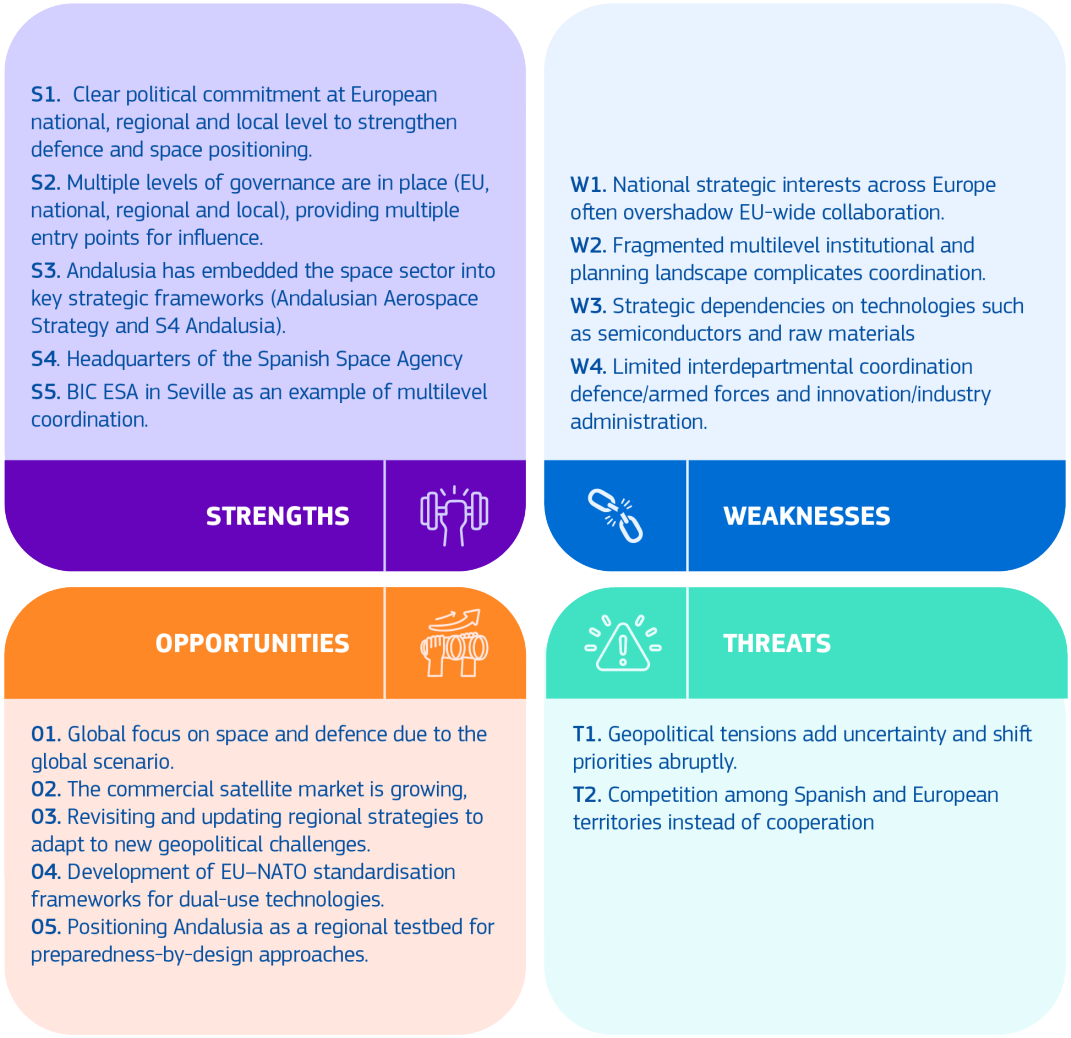
The contributions of experts regarding this subsystem, gathered through interviews and stakeholder consultations, can be synthesised as follows:

- Defence and space sectors are under strong political focus at all governance levels (Outer EU, EU, national, regional and local).
- National interests and political fragmentation complicate EU-wide cooperation.
- Europe demonstrates recognised strengths in telecommunications, with Germany leading in Earth observation and France/Germany in launchers, though the continent remains fragmented.
- Dependence on semiconductors is a strategic vulnerability, making diversification and European alternatives essential.
- ESA's programmes in human space exploration and space defence require greater funding and a clearer framework for civil–defence collaboration.
- The commercial satellite market is growing, but final-stage commercialisation remains a structural weakness in Europe.

- A more formal and stable European collaboration framework in the space sector is strongly recommended.
- Emerging technologies such as AI and quantum technologies are identified as crucial for Europe's future competitiveness in space and defence.
- Lack of standards for EU–NATO dual-use technologies risks slowing adoption.
- Andalusia has already prioritised space and defence through the Andalusian Aerospace Strategy and S4 Andalusia, showing strong regional political commitment.
- Improved multilevel governance, interdepartmental coordination, and international alignment are key to success.
- Maximizing European Space Agency (ESA) Contracts

Based on the analysis of the Orientation and Planning subsystem, the following SWOT is proposed:

Figure 4. Orientation and Planning subsystem SWOT



Source: Authors based on the interviews to the experts

2.2. Resource mobilisation subsystem

2.2.1. Funding mechanisms

Innovation financing in Andalusia is marked by a comparatively lower reliance on private capital—both debt and equity—than in Spain as a whole. Enterprises in the region rely disproportionately on internal funds, while the public sector, particularly universities and government administrations, provides a dominant share of R&D expenditure. This structural imbalance constrains the capacity of innovation-driven firms to scale up effectively.[37]

The imbalance is especially evident in high-technology sectors, such as aerospace and space, where high technological risk, substantial capital intensity, and long investment horizons exacerbate private investors' reluctance to participate. The S4 Andalucía has already emphasised that business R&D financing in the region remains below the national average, with universities and public administrations continuing to dominate expenditure.

In the case of the space sector, this pattern translates into heavy reliance on European and national public programmes (e.g., ESA contracts, CDTI funding, and the PERTE Aeroespacial-Space Technology Plan), while the mobilisation of Andalusian private capital remains limited.

Against this backdrop, the following sections will examine the funding mechanisms for resource mobilisation that are most relevant and applicable to the Andalusian space innovation ecosystem:

- Subsidies and grants
- Entrepreneurship support and venture capital
- Public Procurement
- Internationalisation
- Future funding framework

R&D&I and productive investment grants

Expert's interviews indicate that, as with most Spanish space companies, Andalusian firms, universities, and technology centres are active participants in project calls launched by the European Space Agency (ESA) and Horizon Europe, especially under Cluster 4: Digital, Industry and Space. Several Andalusian companies—such as Integrasys—have also been awarded in space-related calls under the European Defence Fund.

The Spanish Space Agency (AEE) has launched the National Space Technology Plan (Plan Tecnológico Espacial, PTE) as its main instrument to strengthen the country's industrial and scientific base in the space sector. The PTE forms part of the Aerospace PERTE, one of the flagship initiatives of Spain's Recovery and Resilience Mechanism (RRM), with a strong emphasis on applied research and development. In 2024, Andalusia accounted for 13% of the applications submitted to the PTE, ranking third after Madrid (39%) and Catalonia. In terms of project funding, Andalusia

[37] Informe final Evaluación ex ante competitividad pymes. Programa Andalucía FEDER 2021-2027.pdf

received 9% of the total budget, compared with 36% for Madrid, 23% for Catalonia, and 12% for the Basque Country[38].

At the national level, prior to the creation of the Spanish Space Agency—and subsequently in a complementary role—the Centre for the Development of Industrial Technology (CDTI) [39], a public business entity under the Ministry of Science, Innovation and Universities, has played a major role supporting the Space business sector innovation. CDTI's resources are drawn from the Spanish national budget, and it also manages programmes funded by the European Regional Development Fund (ERDF) Operational Programme for Spain, the Recovery and Resilience Mechanism (RRM), among others.

In Spain, potential synergies exist between ERDF or RRM and the co-funding mechanisms of the European Space Agency (ESA) are being explored. ERDF resources can support early-stage activities—such as technology maturation, testing infrastructure, business incubation and public procurement of innovation—that complement ESA's co-funded programmes in upstream and downstream space applications. Strengthening coordination between regional operational programmes and ESA initiatives could therefore enhance Spain's industrial participation, leverage national contributions to ESA, and maximise the impact of EU cohesion policy on the national space ecosystem.

Figure 6 illustrates three levels of grant-based funding support—European, national, and Andalusian—intended as descriptive rather than exhaustive overview. At the national and regional levels, instruments funded by ERDF and RRF are especially relevant, as foreseen in the Andalusian Space Strategy, both at the national level (Spanish Space Agency, CDTI and Spanish Research Agency) and the regional level (Andalucía TRADE, Regional Ministry of Industry and Energy, Regional Ministry of University, Research and Innovation).

e Spanish Ministry of Defence manages COINCIDENTE (Cooperation in Scientific Research and Development in Strategic Technologies), a programme focused on dual-use and defence projects. Its goal is to leverage civil technologies to generate innovative solutions of interest to the defence sector. Projects must deliver a demonstrator with military functionality and represent a significant technological innovation that addresses an existing or potential defence need.

At the regional level and particularly targeted to reinforce regional capacities in the space sector, is the call for projects launched in 2025 by the Regional Ministry of Industry and energy based on Incentives for industry participation in Andalusia in the life cycle of new satellite systems[40], funded by ERDF-Andalusia Operational Programme. with a total budget of €16 million.

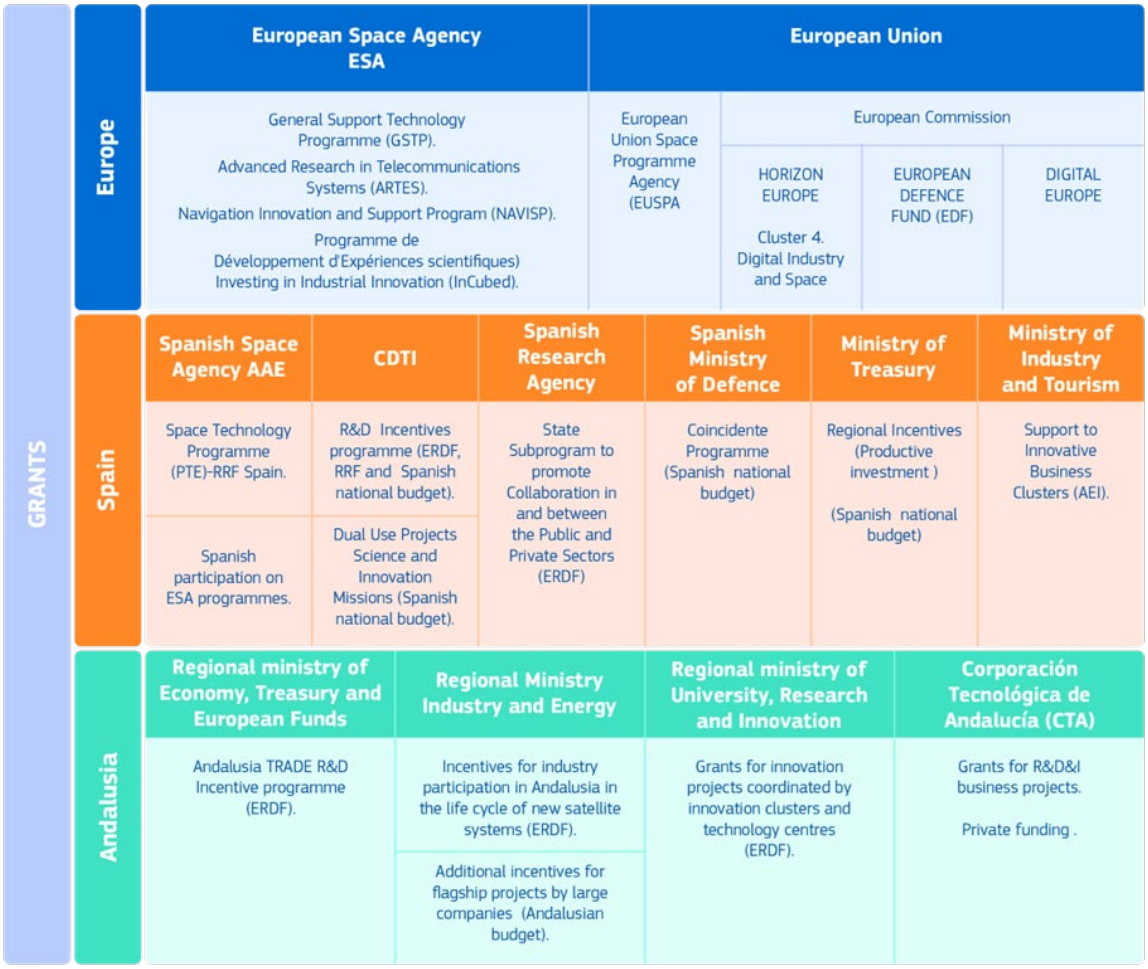
It's important to highlight due to its relevance for the Andalusian space ecosystem, the incentive mechanisms supporting cooperation and clusters. The Spanish Ministry of Industry and Tourism has recognised and awarded the Cluster Andalusia Aerospace. In July 2025 the Regional Ministry of University, Research and Innovation has launched a call for grants amounting to €31,68 million coming from the ERDF Andalusia OP 2021-202741 aiming at consolidating innovation clusters and

[38] Source Spanish Space Agency
[39] Home | CDTI
[40] Incentivos para la participación de la industria en Andalucía en el ciclo de vida de nuevos sistemas satelitales - Junta de Andalucía
[41] https://www.juntadeandalucia.es/presidencia/portavoz/economiayempleo/206827/ayudas/clusteresdeinnovacion/centrostecnologicos/proyectos/InnovAndalucia/Agentes/SistemaAndaluzdelConocimiento/ConsejeriadeUniversidad/InvestigacioneInnovacion

supporting innovation projects coordinated by these entities, projects developed by technological centres, and the acquisition of equipment to enhance their research and innovation infrastructures.

In addition, private resources also support business innovation, notably those managed by Corporación Tecnológica de Andalucía (CTA)[42]—a private foundation with 183 member companies and two decades of experience, originally promoted by the Regional Government of Andalusia and dedicated to fostering technological innovation.

Figure 5. Funding mobilisation R&D&I and productive investment grants



Source: Own authors based on interviews to the experts

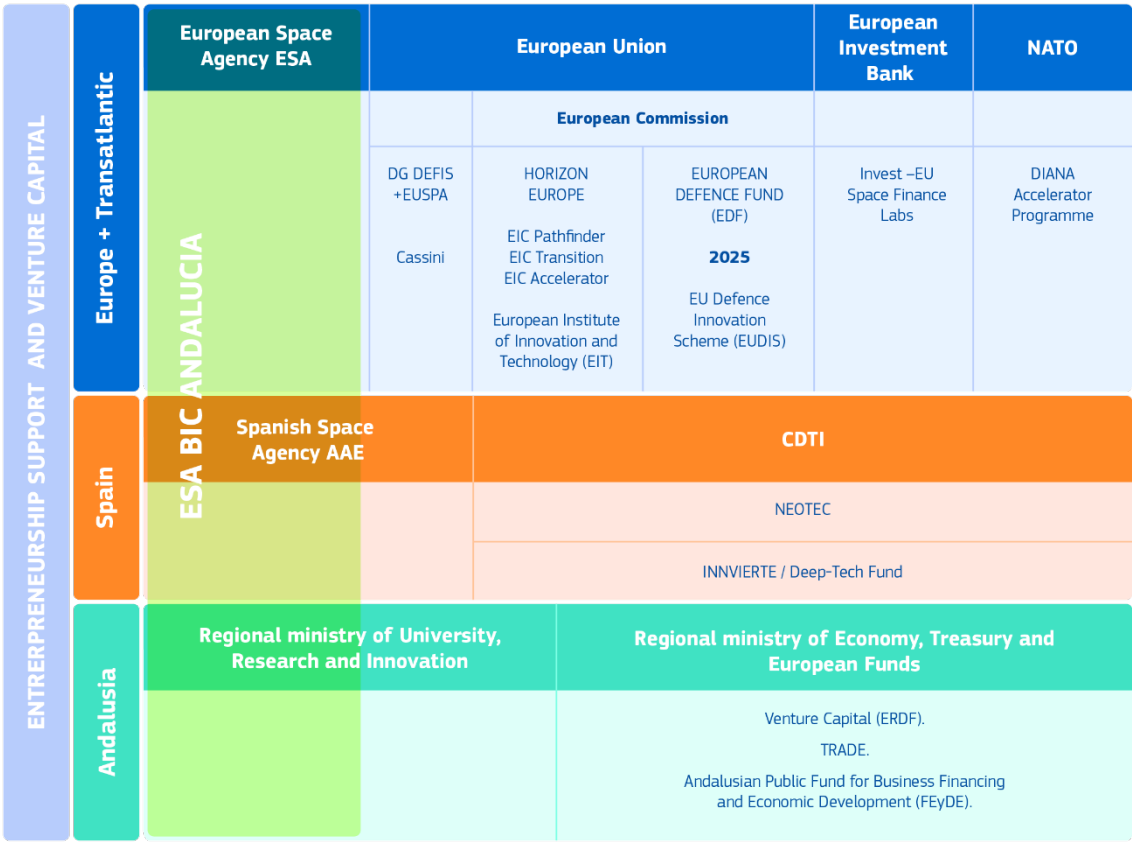
[42] https://www.corporaciontecnologica.com

Supporting entrepreneurship and venture capital

Figure 6 illustrates the tools and resources mobilised to support entrepreneurship and venture capital of interest for the space domain, intended as a descriptive rather than exhaustive overview. These operate at three levels, including intergovernmental and transatlantic initiatives such as NATO's DIANA accelerator programme, the programmes under Horizon Europe (Pathfinder and Accelerator) and under Invest EU. For Andalusia, however, the most significant instrument is the newly established ESA Business Incubation Centre Andalusia

ESA BIC Andalusia is part of the European Space Agency's network of Business Incubation Centres. The ESA BICs are dedicated to turning space-related ideas into successful businesses. The Andalusian centre will provide start-ups with access to ESA expertise, funding opportunities, and a dynamic community of space entrepreneurs. It is funded by the Spanish Space Agency and co-funded by the Regional Government of Andalusia, with additional support from the Seville City Council. Through regular calls for projects, the first one launched on 14 August 2025, ESA BIC Andalusia selects innovative start-ups that use space technologies or apply them to new products and services.

Figure 6. Resources mobilisation. Venture capital and entrepreneurship support



Source: Own elaboration based on the interviews to the experts

Public Procurements

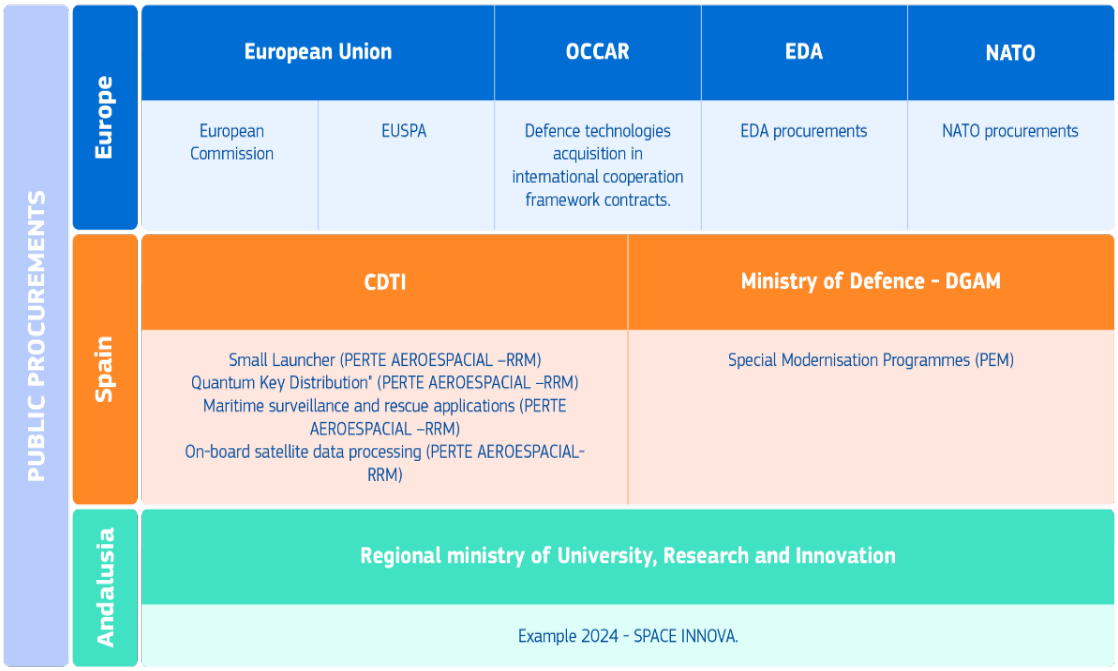
One of the instruments that acts as a powerful demand-creation mechanism—highly valued by companies—is public procurement, particularly public procurement of innovation (PPI). In the case of

European and international institutions, procurement mechanisms are especially relevant. At the Spanish level, the CDTI has traditionally been the main driver of innovation-oriented public procurement programmes. In the space sector, its actions have been particularly significant under the Aerospace PERTE, funded through the Recovery and Resilience Mechanism, with projects such as maritime surveillance and rescue applications, on-board satellite data processing, and quantum key distribution technologies.

In Andalusia, the *SPACE Innova* initiative launched by the Regional Ministry of University, Research and Innovation has been particularly targeted at the preparedness use of space technologies. The programme seeks to identify innovative space-based solutions for applications in emergency response, firefighting, environmental management, and statistical or cartographic analysis. It has a budget of €20 million, co-financed through the Andalusia ERDF 2021–2027 Programme. (See figure 7)

Taking into consideration the new window of opportunity in defence , we highlight the procurement procedures for the Ministry of Defence's Special Modernisation Programmes (PEM), as referred to in the Technological Plan for Security and Defence, are primarily governed by Law 9/2017 on Public Sector Contracts (LCSP), with specific provisions applying to defence and security procurement under Directive 2009/81/EC. This framework allows for special procedures when national security is at stake. The responsible authority is the Directorate General of Armament and Material (DGAM) within the Ministry of Defence. The contracting modalities include, open or restricted procedures, applied when there are no confidentiality or security restrictions; negotiated procedures without publicity, frequently used in strategic programmes, enabling the selection of national prime contractors (e.g. Airbus DS, Indra, Navantia) and framework contracts and international cooperation agreements, often channelled through OCCAR (Organisation for Joint Armament Cooperation) or via European programmes such as the EDF (European Defence Fund) and PESCO (Permanent Structured Cooperation).

Figure 7 Resource mobilisation. Public Procurements



Source: Authors based on the interviews to the experts

Internationalisation

Internationalisation has been a central pillar of public support for Andalusia's space sector, aimed at enhancing the global visibility and competitiveness of regional companies. Between 2020 and 2024, the Andalusian region implemented a range of initiatives to promote the internationalisation of Andalusian companies. From 2021 to 2024, under the leadership of Andalucía TRADE, eight trade missions were organised with the participation of over thirty companies, alongside nine technical information days that engaged more than seventy businesses. A total of €476,730 was allocated to support participation in international trade fairs, benefiting more than seventy-eight companies. Additionally, six inward missions were carried out with a dedicated budget of €55,719 involving over 104 companies.

One of the most important events held in Andalusia is ADM in Sevilla — Aerospace & Defence Meetings. ADM Sevilla is a flagship annual event organised by the Andalusian Regional Government, Andalucía TRADE, in collaboration with BCI Aerospace, aimed at positioning Andalusia as a key hub for aerospace and defence in Spain.

The region also supported twenty individual internationalisation projects and provided consultancy services for two companies aiming to establish a presence abroad. Nineteen companies took part in market prospecting activities, and twenty-two prospective visits were organised, including the involvement of three aerospace firms. Furthermore, three editions of the Aerospace and Defence Meeting were held in 2021, 2022, and 2024, with a combined budget of €449,174[43]

Future funding framework

Looking ahead to the Multiannual Financial Framework (MFF) 2028–2035[44], the European Commission's proposal of 16 July 2025 marks a significant shift in Europe's budgetary priorities, with defence and space budgets increasing five-fold to €131 billion—the largest growth line in the entire budget. For space and defence, the MFF provides targeted increases for capability development, industrial cooperation, and flagship programmes such as Galileo, Copernicus, and IRIS^2, alongside new initiatives in Earth Observation Governmental Services, Access to Space, Space Commercialisation, and dual-use technology innovation.[45]

The MFF proposal also introduces major governance changes to cohesion policy by concentrating the allocation and management of cohesion funds at the national level. Member States and regions will have the possibility to support on a voluntary basis and according to the regional needs and priorities, defence-related projects in their national and regional partnership plans. Under this model, regions would need to negotiate with their central governments to secure the inclusion of funding for securing regional industry support measures—such as space and defence projects—within national partnership plans.[46] While this centralisation could improve coordination and help eliminate overlaps in policy areas that are primarily state competences, it remains to be seen how much room regions will retain to design strategies tailored to their specific territorial strengths and innovation ecosystems.

[43] Monitoring report of the Andalusian Aerospace Strategy 2027. Andalucia TRADE
[44] EU budget 2028-2034
[45] STATEMENT_25_1861_EN.pdf
[46] EU budget 2028-2034

2.2.2. Expert Insights and SWOT

The contributions of experts regarding this subsystem, gathered through interviews and stakeholder consultations, can be synthesised as follows:

- Due to the current geopolitical context, the defence and space sectors are the focus of ambitious programmes at transnational, European, national, and regional levels, attracting unprecedented levels of investment. However, the institutional and funding landscape remains highly complex and fragmented.
- Although there is coordination between ESA, EUSPA, DG RTD (Horizon Europe), the Spanish Space Agency, and CDTI, the multiplicity of requirements continues to hinder the creation of effective synergies.
- SMEs and startups play an increasingly relevant role, yet face barriers in accessing EU funds due to bureaucratic complexity. Simplification of procedures would foster wider participation.
- Andalusian participation in these programmes remains limited, constrained by a lack of awareness, administrative complexity, and capacity gaps. These challenges are even more acute in defence-related programmes managed by NATO and the Spanish Ministry of Defence, where strict standards and certification requirements constitute significant barriers to entry.
- The Andalusian Regional Government is strongly committed to supporting the space and defence sectors, as demonstrated by the adoption of the Andalusian Aerospace Strategy. It is also designing tailored incentives to strengthen regional capabilities, attract investment, encourage participation in consortia, promote cooperation within the cluster, and foster the internationalisation of Andalusian companies.
- Instruments such as pre-commercial procurement could play a key role in developing markets for innovative solutions to address regional challenges. However, experts highlighted that administrative burdens in the design and implementation of public procurement processes remain a significant obstacle to the mainstream adoption of these powerful demand-side innovation tools.

Based on the analysis of the resource mobilisation, the following SWOT framework is proposed:

Figure 8. Resources mobilisation. Subsystem SWOT



Source: Own elaboration based on the stakeholder's consultation

2.3. Production Subsystem

Guided by the (adapted to REGDUALOSA) POINT methodology, this subsystem (see Figure 9) will be examined across the following domains:

- Manufacturers
- Research and Science producer
- Human Resources and Skills Development
- Industrial property
- Research & Test Facilities
- Key projects in the Andalusian space ecosystem
- Clusters and Technological Parks
- Instruments for the alignment with national and EU space and defence agendas

Following the Andalusian space sector map 2022[47], the total number of entities in the region with experience in the space sector reached 196. While companies form a significant proportion of the ecosystem numerically, their levels of engagement and specialisation within the space sector may vary considerably.

2.3.1. Manufacturers

The Andalusian space industry is an emergent domain, characterised by the presence of technology providers and specialised manufacturers. It is composed largely of SMEs operating at different stages of maturity, from early-stage start-ups to well-established firms, although many of them do not have their decision-making bodies located in Andalusia. Airbus Defence and Space maintains a strong defence aviation presence in the region; however, it does not possess space integration capacities in Andalusia. As highlighted during stakeholder consultations, the absence of a major integrator is widely perceived by the business sector as a structural weakness.

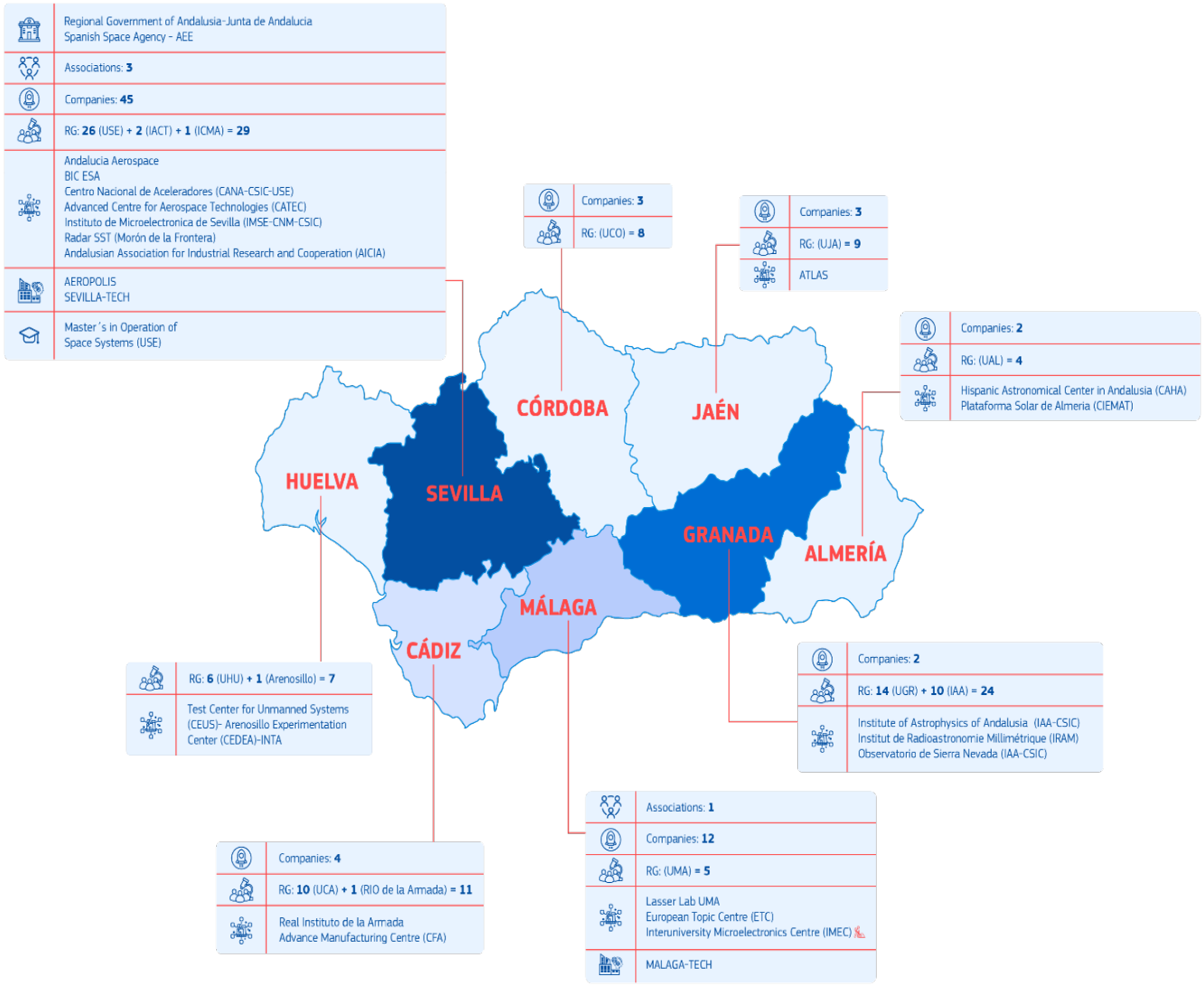
Andalusia's strengths lie particularly in satellite subsystems (eg. solar panels and sensors.,) and payloads, ground segment and signal processing solutions, and engineering and testing services for highly reliable space components. Andalusia also contributes to Earth observation, satellite communications, and navigation programmes at European and international levels. This industrial base is reinforced by R&D collaboration with universities and technology centres. The role of Andalusia Aerospace cluster is pivotal, with the creation of a space commission (task force), currently led by DHV Technologies. The list of the companies with recognised space activity in Andalusia can be found in annex 2 and some of the most relevant space manufacturers located in Andalusia are listed in figure 10.

The Andalusian industrial ecosystem benefits significantly from a dense and advanced network of ICT companies in the region. Although not primarily focused on space, these companies enhance regional capabilities through advanced defence and communications technologies. Collaboration with defence OEMs, such as Navantia, can solidify innovation in naval defence. AYESA contributes expertise in digital transformation and complex engineering systems, while GDELS–Santa Bárbara

47 Mapa Espacio Andalucia 2022_v1-FINAL con anexos.pdf

Sistemas integrates strategic land-defence manufacturing into regional supply chains. Additionally, Escribano M&E and Ghenova, both specialists in naval platforms, defence systems, and infrastructure, provide crucial contributions to high-tech autonomous defence innovation. Alongside other firms throughout the region with robust innovation potential, they collectively complement and strengthen the regional innovation system, supporting Andalusia's emerging space and dual-use capabilities.

Figure 9 Andalusian production Subsystem



Source: Andalusian Space map 2022-Andalusia TRADE and authors

Figure 10. Andalusian most relevant space manufacturers

ALTER TECHNOLOGY (TÜV NORD Group)

- **Focus:** Engineering, procurement, testing, and qualification of electronic components and systems for space.
- **Capabilities:** Radiation and extreme environment testing, failure and constructional analysis, acceptance tests, nanosatellite and CubeSat payload design, assembly, integration and testing.
- **Relevance:** A key European reference in component reliability, participating in ESA programmes and multiple international missions.

DHV TECHNOLOGIES

- **Focus:** Design and manufacture of solar panels and power systems for satellites.
- **Capabilities:** Solar arrays for CubeSats and small satellites, power modules, batteries, thermal insulation, and deployable panel solutions.
- **Relevance:** International footprint with >140 satellites in orbit carrying DHV systems.

GMV

- **Focus:** Integrated solutions covering the whole lifecycle of space missions.
- **Capabilities:** Navigation and PNT, telecommunications, Earth observation, robotics, scientific exploration, and space security.
- **Relevance**: One of Europe's leading space engineering companies; long-standing contracts with ESA and participation in flagship programmes (Galileo, Copernicus).

INTEGRASYS

- **Focus:** Satellite spectrum monitoring and ground segment systems.
- **Capabilities:** Software for carrier/spectrum monitoring, VSAT auto-commissioning, interference detection, network maintenance automation, satellite simulation tools.
- **Relevance:** Recognised innovator in spectrum monitoring, ESA ARTES and EDF programme participation.

SOLAR MEMS

- **Focus:** MEMS-based solar sensors for space and high-tech markets.
- **Capabilities:** High-precision sun sensors, ADCS technologies, ground support equipment, custom designs and mission management for small satellites.
- **Relevance:** Over 1,500 units in flight across 40+ countries; supplier for OneWeb's mega-constellation.

OESIA Group (Tecnobit)

- **Focus:** On-board electronics for space missions and avionics-derived solutions.
- **Capabilities:** Control and power units for antennas, processors and power supplies hardened against radiation, on-board computers.
- **Relevance:** Strategic defence company in Spain, applying defence electronics know-how to space missions (e.g., Galileo PRS service).

INDRA

- **Focus:** Defence, security, and space communications solutions.
- **Capabilities:** Satellite communications terminals and networks, ground control segment, SATCOM "on the move" solutions, participation in Galileo.
- **Relevance:** Spain's leading technology company in space-related defence communications and a trusted partner of ESA and EUTELSAT. It has been announced in July 2025 that INDRA GROUP will double its activity in Andalusia and expand its workforce to 3.700 professionals.

MADES

- **Focus:** Defence, security, and space communications solutions and electronics manufacturing
- **Capabilities:** Production of high-precision electronic assemblies and subsystems for aerospace and defence markets; optical systems via Raytheon ELCAN (e.g. space-grade EO/IR subsystems).
- **Relevance:** Based in Málaga; key supplier to tier-one aerospace and defense customers. Recognised for excellence in electronics

AYTANA

- **Focus:** Defense, Aerospace, Hydrogen, Big Science Facilities
- **Capabilities:** Spanish spin-off of the Cox Group, inheriting over 20 years of experience in defense and hydrogen technologies and 12 years in aerospace and large scientific facilities
- **Relevance:** involved in prominent programmes such as Ariane, ESA initiatives, and Vega-C.

Source: Own elaboration

2.3.2. Research & Science

The 2022 Andalusian Space Sector Map already documented dozens of university groups tied to upstream (payloads, satellites, ground segment) and downstream (EO/positioning apps) activities; today, these capabilities are deepening via new chairs, ESA/EU contracts, small-sat projects, and SST/STM collaborations—linking universities directly with the region's companies and infrastructures.

The Andalusian university system plays a central role in supporting the region's space and defence ecosystem through cutting-edge research, infrastructures, and talent generation. The University of Seville stands out with its Aerospace Engineering School, laboratories in structures, propulsion, and avionics, and participation in ESA and Horizon Europe projects. The University of Granada is highly active in astrophysics, satellite data processing, and Earth observation, hosting research groups linked to the Instituto de Astrofísica de Andalucía (IAA-CSIC). The University of Málaga is recognised for its strengths in telecommunications, navigation systems, and artificial intelligence applied to space technologies, and hosts the Andalusian Institute for Data Science and Computational Intelligence. The University of Cádiz has specialised capacities in naval engineering, new materials, and energy systems, relevant for dual-use maritime and space applications. The University of Jaén contributes through research in robotics, advanced mechanics, and applied computing, with growing links to aerospace and defence applications.

Andalusian universities play an active role in European university alliances, with the University of Seville (Ulysseus), the University of Granada (Arqus), and the University of Cádiz (SEA-EU) contributing to cross-border knowledge, innovation, and advanced sectors, including connections to aerospace through regional research and technology ecosystems.

Beyond degrees and projects directly related to space and defence, Andalusian universities host strong potential in transversal disciplines that are essential for the sector's competitiveness. Artificial intelligence and data science—with leading groups in Seville, Granada, and Málaga—are applied to satellite data processing and autonomous space vehicles. Software engineering and cybersecurity, present across the regional university system, are critical for mission control systems, ground segment operations, and secure communications. New materials and nanomaterials, with strong research bases in Cádiz, Málaga, and Seville, fuel innovation in lightweight structures, protective coatings, and radiation-resistant components. Added to this are fields such as photonics and microelectronics (with industrial-academic connections to companies like Solar MEMS or Alter Technology), quantum computing, and advanced robotics, which are also part of the Andalusian academic ecosystem. Together, this multidisciplinary knowledge base strengthens the space and defence industries and provides a solid foundation for the development of dual-use technologies in Andalusia. Drawing in the latest news and publications of the Andalusian Universities the following highlights can be presented on the Andalusia research in space & defence (Andalusia) university's contribution. See illustration 11.

As part of the research institutions located in Andalusia, the European Commission's Joint Research Centre is a European body that acts as a connecting and catalysing element for the Andalusian innovation ecosystem, which is why we include as part of the ecosystem itself. It was established in Seville in 1994 and provides socio-economic and techno-economic analysis in support of the design, development, implementation, and monitoring of EU policies. JRC Seville is specialised among others, in the following areas: Circular economy and sustainable industry, Industrial strategy, skills

and technology transfer, Innovation policies and economic impact and algorithmic transparency. Among the JRC projects related to space technologies ESETA, on the development of the European space economy statistics, and REGDUALOSA, the one that frames this report, can be mentioned.

On the other hand, as stated in the diagnosis document of S4 Andalucia, the application of available knowledge and the use of industrial property are still limited, and knowledge transfer between universities, public research organisations, and companies could be strengthened. In addition, some research activity does not yet fully align with the needs of industry, which reduces the effectiveness of public investment in R&D and limits the direct economic impact of scientific output. Addressing these gaps offers significant opportunities to improve collaboration, enhance the transfer of technology, and increase the sector's contribution to regional competitiveness.

2.3.3. Human Resources and Skills Development

The Space Skills Alliance provides Europe-wide analysis and tools for addressing space-sector skills gaps, offering insights that inform national and regional workforce planning. The ESA Academy delivers structured training and practical learning opportunities, complemented in Spain by activities hosted at the European Space Astronomy Centre (ESAC) near Madrid and by the European Space Education Resource Office (ESERO) Spain, which operates one of its centres in Granada. In parallel, the National Institute for Aerospace Technology (INTA) strengthens the national talent pipeline through recurring paid training scholarships and technical capacity-building programmes.

The development of Andalusia's space sector relies fundamentally on the availability of a highly skilled workforce. To date, the region's educational offer has been primarily shaped by aerospace engineering programmes—traditionally focused on aeronautics—delivered at the University of Seville and the University of Cádiz, where specific modules are dedicated to space technologies.

A major step forward is the creation of the master's degree in Operation of Space Systems (MOSE) at the University of Seville's Higher Technical School of Engineering, due to commence in September 2025. This will be the first specialised master's in space technologies in Andalusia, taught fully in English, designed to integrate skills across aeronautics, telecommunications, control systems, ground communication, and satellite data transmission.

Complementing this, microcredentials such as "Electronic Technology for the Space Sector" (University of Seville) are creating new pathways for early exposure to space-specific competencies. Microcredentials represent a fundamental tool for the Andalusian space and defence industries, offering flexible and modular training that supports professional re-adaptation, lifelong learning, and specialisation in emerging domains. They allow professionals from related fields to reskill rapidly for space applications, ensure continuous updating of competences in fast-evolving areas

At the vocational training level, Andalusia is progressively aligning programmes with the needs of the aerospace and space value chain. This includes professional qualifications in aeronautical maintenance, avionics, composite materials, and industrial automation, offered through Andalusian vocational schools with strong ties to the aerospace industry.

Beyond strictly space-focused training, Andalusia hosts advanced programmes in fields that are key enablers for space and defence technologies:

Figure 11 Andalusian Universities' space activities

University of Seville (US)

- Education & talent: New Master's in Operation of Space Systems (MOSE), 100% in English, bridging aerospace and telecoms control/communications for spacecraft operations. Notable directions: space surveillance/STM, onboard electronics & robust computing, robotics for space systems.
- Strategic capability: Indra–US Chair in Space Surveillance (STM focus) working with AEE, ESA, INTA, COVE and ROA to advance space surveillance R&D and skills.
- Research groups & EU projects: Robotics, Vision & Control (GRVC, TEP151); validation of rad-hard FPGA & tools (TIC192); participation in astronomy e-infrastructure networks

University of Granada (UGR) & IAA-CSIC (Granada)

- Flagship institute: IAA-CSIC—major European centre for astrophysics & instrumentation; participation in Solar Orbiter and JUICE missions; leverages Calar Alto Observatory as tech test-bed. IAA CSIC. Notable directions: space science payloads/instrumentation, space weather/planetary data, observatory operations.
- Regional infrastructure: Junta funding supports ongoing Calar Alto operations and new instrumentation
- Programmes & groups: Multiple astronomy/space groups (e.g., ARAE high-energy & robotic astrophysics; planetary atmospheres; radioastronomy) with EU e-infrastructure links (ASTERICS, ESFRI clusters).

University of Málaga (UMA)

- Space surveillance (SST/SSA): ESA contract with IAA-CSIC & UMA to design a cooperative observation/control system for Spain's S3T/EUSST—leveraging the global BOOTES robotic telescope network and UMA's Space Robotics Lab. Notable directions: SST/STM software & autonomy, satcom antennas & D2D, radiation-tolerant photovoltaics.
- Space communications & 5G/Direct-to-device: New European hub for mobile broadband (space/terrestrial) with Vodafone and AST SpaceMobile, hosted by UMA. Uma
- Space tech R&D: Work on radiation-hard solar cells (EIC Pathfinder ZEUS); metasurface leaky-wave satcom antennas; space-weather forecasting.

University of Cádiz (UCA)

- Small satellites: UCAnFly educational nanosatellite under ESA's Fly Your Satellite! programme—testing emerging tech (e.g., for gravitational-wave detection concepts). Notable directions: cubesat education & payload prototyping, positioning & cartography, aerospace systems.
- Skills pipeline: BSc in Aerospace Engineering (Puerto Real) aligned with maritime/aerospace synergies.
- Research lines: Lab of Astronomy, Geodesy & Cartography; signal/systems engineering.

University of Jaén (UJA)

- UAS & flight-test ecosystem: Collaboration framework with FADA-CATEC and the ATLAS flight-test centres (Villacarrillo/Beas de Segura) Notable directions: UAS testing & autonomy, geomatics/photogrammetry, fluid dynamics for aero/EO.
- Research groups: Cartographic engineering (TEP164), photogrammetry/topometry (TEP213), fluid mechanics (TEP235) supporting EO/remote-sensing and aerodynamics.

University of Córdoba (UCO)

- Education & talent: Strong engineering and ICT programmes with growing specialisation in logistics, cloud computing and defence digitalisation. Participation in national initiatives such as the Spanish Army's Logistics Base project, reinforcing dual-use competencies in digital transformation.
- Strategic capability: Close collaboration with defence and industrial stakeholders on logistics systems, AI applications, and sustainable engineering.
- Research groups & EU projects: Notable groups include Computational Simulation in Engineering (TEP235) and Artificial Intelligence & Knowledge Engineering (TIC169).

Source: Own elaboration

- Mechatronics and Robotics: The joint Bachelor's in Electronic Engineering, Robotics, and Mechatronics offered by the Universities of Seville and Málaga (Andalucía TECH Campus) provides specialisations in automation, electronic instrumentation, and mechatronic vehicle systems. UMA also leads research in robotics and autonomy with applications in aerospace and space systems.
- Cybersecurity: Universities such as Seville, Málaga, and Granada offer master's degrees and research programmes in cybersecurity, a field of growing relevance for satellite systems, ground infrastructures, and dual-use applications.
- Artificial Intelligence: Andalusia's universities—particularly the University of Granada—are leaders in AI research, with direct applications in data processing, Earth observation, satellite autonomy, and defence systems.
- Quantum Technologies: Early research and training initiatives, particularly in Málaga[48] and Granada, are fostering expertise in quantum communications and quantum sensing, both of which are expected to transform secure satellite communications and navigation.

Although Andalucía's labour market is highly attractive from a production cost perspective, this advantage also carries the risk of outflow of talent toward better-paid opportunities in other regions and continent. The rapid growth in demand for qualified professionals in space-related technologies highlights the urgent need to both increase the number of graduates in STEM disciplines and support the reskilling of professionals from less in-demand fields.

At the same time, there is a growing requirement for specialised vocational and specialised profiles, an area where training opportunities remain underdeveloped. In this regard, the collaborative efforts between the Aerospace sector and the Regional Government of Andalucía have resulted in a partnership aimed at addressing these challenges and strengthening the region's human capital base[49].

As recently announced in the EU Union of Skills Communication[50], the EU will develop sector-specific skills and intelligence, and prioritise creating, attracting and retaining top talent by offering attractive career paths and opportunities for growth within the EU, as well as supporting reskilling schemes. The space specific skills intelligence methodology will, inter alia, rely on ongoing Erasmus Blueprints & Pact partnerships and will feed into the new European Skills Intelligence Observatory.

The Union of Skills was accompanied by the launch of the STEM Education Strategic Plan, outlining various measures for advancing STEM education and training to increase talent across the EU. The Plan includes ambitious targets to boost enrolment in STEM fields in vocational education and training (VET) and higher education by 2030, with a particular focus on attracting more girls and women to those areas.

[48] Integrated quantum photonics | Photonics & RF Research Lab

[49] Empleo y el sector aeroespacial firman un protocolo para formar a profesionales en Cádiz, Sevilla y Málaga - Portavoz del Gobierno Andaluz

[50] COM(2025) 90

2.3.4. Industrial property in Andalusia's space Sector

The Andalusian government—chiefly through Andalucía TRADE—plays a pivotal role in streamlining processes for patent generation and industrial protection for entrepreneurs, SMEs, research centres, and universities active in the space and dual-use sectors. These institutional efforts are particularly significant considering the documented industrial property outputs in the *Andalusian space sector map 2018* and 2022.

Key Examples

- FADA–CATEC (Seville): *Compact system for torque generation and control* (class B64G1).
- University of Cádiz (TIC213 – Naval Signals, Systems and Communications Group): *Demonstrator of satellite radiocommunications concepts for equatorial orbits, with multiple applications in higher education*.
- University of Huelva (TEP994 – Multidisciplinary Applied Engineering): *High-energy particle detection device*.
- University of Seville (TIC21/TIC231, among others): *Device, system, and method for energy-efficient satellite-based location and tracking*.
- Individual inventors (Málaga, Cádiz): Patents on *satellite-based personal security and position control systems*, *satellite-enabled cardiological monitoring systems*, and *radiotelephones with GPS positioning*.
- University of Jaén – RNM282 Microgeodesia Group: *Procedure for estimating the position of a receiver using GNSS satellites* (ES2444513 A2).
- University of Seville (various TIC groups): *Device, system, and method for energy-efficient satellite-based location and tracking*

2.3.5. Research & Test Facilities

Andalusia hosts a dense and diverse network of research and test facilities that provide the technological backbone for its aerospace, space, and defence ecosystem. These infrastructures, ranging from astrophysics observatories and microelectronics centres to advanced manufacturing hubs and unmanned systems test ranges, reflect the region's capacity to combine fundamental research with applied industrial innovation.

Distributed across all major Andalusian provinces, they form a geographically balanced system that links universities, public research organisations, and industrial clusters.

Andalusian Astrophysics Institute (IAA-CSIC) and Calar Alto Observatory Granada-Almeria

The Andalusian Institute of Astrophysics (IAA–CSIC) in Granada is a leading centre for astrophysical research and instrumentation. It collaborates on significant Earth- and space-based projects and operates Calar Alto Observatory in Almería, jointly managed with the Junta de Andalucía, which features three telescopes (1.23 m, 2.2 m, and 3.5 m apertures) plus an additional 1.5 m telescope under Madrid's observatory control.

Centre for Advanced Aerospace Technologies (CATEC) -ATLAS – Space, Advanced Manufacturing & Flight Testing- Seville-Jaen

The Centre for Advanced Aerospace Technologies (CATEC), managed by the Andalusian Foundation for Aerospace Development (FADA) and based in Seville, is a leading player in aerospace innovation with growing impact in the space and defence sectors.

CATEC has contributed to European and national space programmes in collaboration with ESA, the Spanish Space Agency, Airbus, GMV, Alter Technology and Indra. Its main areas include on-board systems and avionics, robotics and autonomy for space operations, qualification of materials and additive manufacturing (AM) for spacecraft structures, and ground support equipment for satellites.

It is recognised nationally as a pioneer in additive manufacturing, developing and qualifying metal AM parts for satellites and launchers, and transferring digitalised production and robotic inspection technologies into industrial supply chains alongside Airbus, Navantia and Andalusian SMEs.

Beyond Seville, FADA-CATEC operates the ATLAS Flight Centre in Villacarrillo (Jaén), a unique European facility for UAS and HAPS (High-Altitude Pseudo-Satellites) testing. ATLAS provides dedicated segregated airspace, advanced infrastructure, and proven experience in stratospheric platforms that enable Earth observation, communications, and security applications.

INTA CEUS and CEDEA Huelva

The National Institute for Aerospace Technology (INTA), operates two key infrastructures in Huelva: the El Arenosillo Experimentation Centre (CEDEA) and its extension, the Test Center for Unmanned Systems (CEUS). Together, they form one of Europe's most advanced facilities for testing unmanned aerial systems (UAS) and supporting space-related activities. CEUS provides a segregated one-million-hectare no-fly zone, complemented by CEDEA's radar, optronic, telemetry and communications systems.

CEUS/CEDEA hosted the launch of Miura 1 in October 2023, the first Spanish reusable suborbital rocket developed by PLD Space. This milestone represented a breakthrough for Europe's private space sector and underscored the versatility of the Andalusian test range for both unmanned aerial systems and space access.

Commissioned in 2024 with joint support from INTA, CDTI, and the Government of Andalusia, the centre offers unique conditions for the qualification and certification of large unmanned aerial platforms. Early operations included the Milano UAS, a tactical long-endurance drone that carried out certification test flights under STANAG-4671 protocols.

National Centre for Accelerators (CNA)-Seville

The National Centre for Accelerators (CNA) is a Unique Scientific-Technical Facility (ICTS) jointly managed by the University of Seville, the Junta de Andalucía, and CSIC. It houses six key instruments: a 3 MV Tandem Van de Graaff, an 18/9 MeV cyclotron, a 1 MV mass spectrometer, a PET/CT scanner, the MiCaDaS radiocarbon dating system, and a ^60Co irradiator. Applications cover material science, environmental research, nuclear instrumentation, biomedical imaging, and preclinical molecular studies.

UMA Laser Laboratory- Málaga

The Laser Laboratory within the Department of Analytical Chemistry at the University of Málaga, led by Prof. Javier Laserna, specialises in laser-based material analysis and processing. Since 1987, it has engaged in projects across sectors such as microelectronics, cultural heritage, environment, defence, and space, supported by national and regional research funding.

CFA – Centre for Advanced Manufacturing Technologies- Cádiz

Developed in Puerto Real with Navantia's support, the Centre for Innovation in Advanced Manufacturing Technologies (CFA) serves as a bridge for regional manufacturing firms, offering technical support for developing transferable production technologies. It is backed by ERDF funding, amounting to €18.31 million, with 77% already allocated.

Microelectronics Institute (IMSE-CNM)-Seville

A leading microelectronics research and development centre, jointly operated by CSIC and the University of Seville, is located within Sevilla TechPark at Cartuja. The centre specialises in the design, testing, and characterisation of CMOS (Complementary Metal–Oxide–Semiconductor) technologies, widely used for building analog and mixed-signal integrated circuits, as well as radio-frequency (RF) components and microsystems.

The centre comprises seven state-of-the-art laboratories dedicated to logic, electrical, functional, and thermal characterisation, in addition to an integrated circuit design laboratory equipped with nanometric fabrication tools. It is further reinforced by the USE CHIP Chair in Microelectronics, which focuses on training highly qualified specialists in semiconductor design and contributes to strengthening the regional microelectronics ecosystem.

IFMIF-DONES (International Fusion Materials Irradiation Facility)-Granada

A groundbreaking particle accelerator facility under development in Escúzar, Granada, aimed at simulating neutron irradiation conditions akin to future fusion reactors.

IMEC – Semiconductor R&D Centre (Málaga TechPark)

A forthcoming IMEC[51] centre within Málaga TechPark will be dedicated to microelectronics R&D, including experimentation with new materials and chip prototyping. Supported by land transfers and funding from the Andalusian Government, this initiative positions Andalusia as a global innovator in smart semiconductors—critical for aerospace, space, and defence applications.

Space Surveillance and Tracking (S3T)

Coordinated by the Spanish Space Agency, S3T is a service providing collision avoidance, re-entry prediction, and fragmentation analysis, implemented in cooperation with the European Space Agency and jointly managed by civil and defence authorities. The Spanish Space Surveillance and Tracking Surveillance Radar (S3TSR), It is a low Earth orbit (LEO) surveillance radar operating in the L band. The radar is installed at the Morón de la Frontera air base (Seville).

2.3.6. Key projects in the Andalusian space ecosystem

Over the past decade, Andalusia has gradually strengthened its space ecosystem through participation in European and national programmes, the mobilisation of research centres, and the creation of industrial clusters. This initial phase laid the groundwork for a new stage focused on capacity-building and consolidation. The current portfolio of regional projects—ranging from nanosatellites for agriculture and environmental monitoring, to CubeSats fully designed with Andalusian technology, and innovative subsystems for NewSpace constellations—illustrates how the region is investing in the skills, technologies, and infrastructures needed to position itself as a relevant actor in the European space landscape. Three of these initiatives are presented below, as essential steps in reinforcing Andalusia's industrial and scientific base and in opening pathways for future market integration.

PLATERO[52]

Andalusia successfully launched the *Platero* nanosatellite in 2023 from the Vandenberg Space Force Base (California, USA). This historic initiative was promoted by the Andalusian Ministry of

51 https://www.imec-int.com/en/connect-us/imec-spain

52 Andalucía lanza con éxito el nanosatélite Platero al espacio y muestra su impulso a la Agricultura 5.0 - Noticias - Junta de Andalucía

Agriculture, Water, Fisheries and Rural Development through the Agrarian and Fisheries Management Agency of Andalusia (Agapa), with the support of the European research infrastructure LifeWatch ERIC and the collaboration of the ceiA3 Agrifood Campus of International Excellence**.**

Platero, a 10-kg nanosatellite, forms part of the *SmartFood LifeWatch ERIC* programme[53]. Equipped with a state-of-the-art multispectral camera, it captures Earth observation images and integrates them with data from terrestrial and maritime IoT-based sensors. This capability enables real-time environmental monitoring and provides updated diagnoses of agricultural and marine ecosystems in Andalusia, including the assessment of climate change impacts on farming and fisheries. As Andalusia's first nanosatellite technology project**,** Platero demonstrates the region's capacity to apply space innovation directly to sustainable agriculture and marine resource management**.**

ALPHA³

The Misión Alpha *project*[54] *represents* a milestone as the first satellite entirely designed, built, and launched with Andalusian technology. Coordinated by the OnTech Innovation Cluster, it brings together companies, universities, and research centres to promote a regional "space economy".

The *Alpha³* satellite, scheduled for launch in 2025, is a CubeSat measuring 10 × 30 cm and weighing approximately 1.8 kg. Its payload is designed to measure space radiation and geomagnetic fields, providing valuable data for regional research centres and supporting Andalusia's growing presence in the NewSpace sector. Beyond its scientific goals, *Alpha³* symbolises Andalusia's ambition to consolidate an autonomous industrial base for space systems.

DIGISOLAR

The DIGISOLAR [55]project (*"New Asynchronous Digital Solar Sensor for NewSpace Applications with Event-Based Architecture"*) aims to design and validate a next-generation digital solar sensor for satellite attitude control systems. The project is funded by Spain's Ministry of Industry, Trade and Tourism.

Its innovative asynchronous, event-based architecture seeks to achieve substantial improvements over current state-of-the-art sensors. Once validated, the DIGISOLAR prototype will strengthen Andalusia's industrial leadership in solar sensing technologies for CubeSats and NewSpace constellations, building on the existing expertise of regional companies such as Solar MEMS.

These initiatives show how Andalusia is building the skills and technologies needed to link space with key sectors such as agriculture, marine management, and advanced manufacturing. By bringing together public institutions, research centres, and industry, the region is laying solid foundations for a sustainable space ecosystem.

2.3.7. Clusters and Technological Parks

Andalusia's aerospace and space ecosystem is strongly reinforced by a cluster and three technology parks that act as platforms for cooperation, innovation, and industrial development. These structures not only concentrate business activity but also connect industry with research, education, and public policy, creating the conditions for long-term competitiveness in strategic sectors such as space and defence.

[53] LifeWatch ERIC
[54] misionalpha.eu
[55] https://andaluciaaerospace.com/proyecto/digisolar/

Andalucía Aerospace Cluster

The Andalucía Aerospace Cluster[56] is central to consolidating Andalusia's aerospace and space industries, serving as the principal platform for collaboration, representation, and international reach. Established in 2017, the cluster now encompasses over 148 companies, research institutes, and academic partners.

Andalucía Aerospace drives cross-sector innovation, working alongside rail and naval clusters and pioneering inter-cluster collaboration with adjacent sectors such as logistics and mobility. The cluster's strategic ambition extends across borders through active involvement in the AERIS+ project, funded under the Spain–Portugal Cross-Border Cooperation Programme (POCTEP), which promotes joint aerospace and defence initiatives with Portuguese counterparts. It also participates in international programmes via the European Aerospace Cluster Partnership (EACP), connecting Andalusian capabilities to pan-European networks.

Its operations are firmly supported by the Government of Andalusia through mechanisms such as Andalucía TRADE, ensuring alignment with overarching European (Horizon Europe, Clean Aviation, ESA) and national frameworks.

Aerópolis – Aerospace Technology Park of Andalusia (Seville)

Aerópolis, located in La Rinconada (Seville), is the only specialised aerospace science and technology park in Spain, and constitutes the epicentre of Andalusia's aerospace innovation ecosystem. Conceived as a cluster platform, it brings together over 90 companies and organisations dedicated to aerospace, defence, and advanced engineering. The park integrates SMEs, start-ups, Tier-1 suppliers, and multinational anchors such as Airbus, alongside R&D centres and technology transfer entities.

Aerópolis concentrates more than 60% of Andalusia's aerospace companies, representing around 40% of sectoral turnover and 33% of employment in the regional aerospace industries: The park hosts companies working in aerostructures, space components, avionics, UAVs, composites, advanced manufacturing, and engineering services. It is closely linked with Andalucía Aerospace, University of Seville, US, and technology centres such as CATEC. It also provides incubators and business support services for start-ups and spin-offs.

Sevilla TechPark

Sevilla TechPark, is the leading innovation hub of the city of Seville. Established on the legacy grounds of Expo '92, it hosts a diverse ecosystem of technology-intensive companies, research centres, and knowledge institutions. The park accommodates 575 companies and organisations, employing more than 31,600 professionals, with an estimated annual economic activity exceeding €5.5 billion.

Its scope is broader than Aerópolis, covering ICT, renewable energy, engineering, advanced services, biotechnology, and creative industries, while also maintaining important connections with aerospace and space-related firms and research groups.

[56] Informe del sector aeroespacial andaluz 2024 - Andalucía Aerospace

As the Seville city's principal innovation park, it acts as a cross-sectoral bridge for the aerospace and space ecosystem, linking them with digital technologies, smart systems, and Industry 4.0.

Málaga TechPark

Málaga TechPark, is one of the largest and most dynamic science and technology parks in Spain and Southern Europe. The park hosts more than 715 companies and 27,900 employees, with an innovation-driven economic profile. It has consolidated its leadership in ICT, microelectronics and semiconductors, cybersecurity, connected and autonomous vehicles, Industry 4.0, digital health and biotechnology, smart cities, renewable energies, and advanced engineering consultancy.

The strong presence of microelectronics, semiconductors, and cybersecurity companies, together with multinational R&D centres (Oracle, Accenture, Ericsson, TDK, Huawei, and soon IMEC), makes Málaga TechPark a critical node for dual-use technologies that underpin the space, defence, and security industries. Its ongoing collaboration with the University of Málaga (UMA) further reinforces applied research in AI, robotics, telecommunications, and space-related materials

Taken together, Andalucía Aerospace, Aerópolis, Sevilla TechPark, and Málaga TechPark not only concentrate aerospace and space activity but also connect it with adjacent sectors such as ICT, naval, and rail, while fostering cross-border and European collaborations.

2.3.8. Instruments for the alignment with national and EU space and defence agendas

The Andalusian regional government, together with leading industrial, scientific, and academic actors, actively participates in a wide range of programmes and networks that align with European and national strategies in aerospace, space, and defence technologies.

Enterprise Europe Network (EEN)[57]

Through Andalucía TRADE, the region is integrated into the EEN, which offers advisory services, access to technology transfer opportunities, and facilitates cross-border partnerships for companies and research groups engaged in aerospace and defence.

Since 2015, Andalusia has been hosting the International Forum on Dual-Use Technologies, organised by Andalucía TRADE in the framework of the EEN. This annual forum has contributed to fostering a dynamic ecosystem around dual-use technologies by bringing together SMEs, large corporations, research organizations, and universities to explore collaborative opportunities. The edition 2024 was titled "The Space Economy: Driving Transformation"[58]

European Network of Defence Related Regions ENDR[59]

The European Network of Defence-related Regions (ENDR) was created in 2016 by the European Commission as a platform to foster regional cooperation and strengthen the role of SMEs and clusters in the European defence value chain. Although it is not directly funded by the European Defence Fund (EDF), the ENDR actively supports stakeholders in improving access to EU funding

[57] CESEAND - Nodo Andaluz de la EEN - Andalucía TRADE
[58] Dual-Use Technologies 2024 - Home
[59] European Network Of Defence-Related Regions

opportunities, including the EDF, Horizon Europe, and other programmes. Its main objective is to promote the uptake of defence-related projects, enhance cross-border collaboration, and increase the competitiveness of SMEs.

Andalucía TRADE has been a member of the network since its inception, actively contributing to its initiatives. Notably, the International Forum on Dual-Use Technologies, mentioned above, has been organised in collaboration with the ENDR.

European Aerospace Cluster Partnership (EACP60)

The Andalucía Aerospace Cluster is an active member of EACP, promoting exchange of best practices and joint European projects. This membership strengthens inter-cluster cooperation with regions in Spain (Madrid, Castilla-La Mancha, País Vasco) and with European aerospace hubs in France, Germany, and Portugal.

National Support Mechanisms (ICEX and CDTI)

Andalusian companies are supported by national organisms such as ICEX and CDTI. ICEX Spain provides internationalisation services, training, and visibility in strategic markets. CDTI funding instruments further enable regional firms to participate in ESA, Horizon Europe, and European Defence Fund projects.

SME4SPACE

Two Andalusian space companies are active members, Solarmens and DHV, in SME4SPACE which is the Representative Organisation of SMEs in Europe's Space Agency.

2.3.9. Expert Insights and SWOT

The analysis of the Production Subsystem is based on insights gathered through expert interviews and stakeholder consultations. This subsystem is examined across the domains of manufacturers, research and science, industrial property, research and test facilities, human resources and skills development, key projects, clusters and technological parks, and alignment with national and EU space and defence agendas.

The consultation process revealed both the progress made and the challenges still to be addressed in consolidating Andalusia's space industry. Stakeholders emphasised the importance of scaling up companies, fostering stronger cooperation across firms, clusters, and institutions, and maximising cross-technology synergies with UAVs, cybersecurity, AI, and other dual-use fields.

Main Take-aways from interviews and stakeholder consultations

- Emerging sector: The Andalusian space domain is growing within a consolidated aerospace industry. However, most firms are small and undersized, limiting investment capacity, innovation, and talent attraction. Scaling up and strengthening cooperation between companies is a priority.

[60] EACP | EUROPEAN AEROSPACE CLUSTER PARTNERSHIP

- Cross-technology convergence: Andalusia benefits from complementary strengths in ICT, UAVs, cybersecurity, AI, robotics Exploiting these synergies can accelerate dual-use innovation and reinforce the region's critical mass
- Defence integration: Anchoring the ecosystem through cooperation with space integrators (Airbus D&S, SENER, Thales Alenia, OHB) and defence primes (Navantia Sistemas, Escribano, General Dynamics Santa Bárbara, Rheinmetall) is critical. This will enable SMEs to access higher segments of the value chain.
- Research and test infrastructures as assets: Facilities such as CATEC and its ATLAS centre, CEUS/CEDEA, IMSE-CNM, CNA, UMA Laser Lab, and IFMIF-DONES provide unique testing ranges and advanced laboratories. These infrastructures anchor Andalusia in international research networks but must be complemented by large-scale industrial spaces (e.g. Aerospace District of Seville) to enable scaling from R&D to full industrialisation.
- Access and visibility challenges: Andalusian companies face difficulties accessing EU funding due to complex procedures and intense competition. Simplification of funding rules and targeted instruments for SMEs are required.
- At European level, programmes such as EU Defence Innovation Scheme (EUDIS) -within EDF- promoted by DG DEFIS of the European Commission offers a set of instruments for small and medium-sized enterprises (SMEs), including start-ups, and other non-traditional players in defence industrial ecosystem to ensure more opportunities to access the European Defence Fund.
- Human capital challenges: Andalusia has robust academic foundations (aerospace, telecommunications, and electronics), complemented by emerging space-specific programmes (e.g. Master in Space Systems Operation, microcredentials in space electronics). However, talent retention remains a challenge, requiring more competitive opportunities and lifelong training for technical profiles.
- Administrative and financial barriers: Bureaucracy at the regional and EU levels continues to hinder SME competitiveness. Tailored funding instruments for startups, SMEs, and mid-caps are needed to ensure growth.
- Cluster dynamics: Clusters such as Andalucía Aerospace and OnTech Innovation, together with technology parks (Aerópolis, Sevilla TechPark, Málaga TechPark), form a strong collaborative base. Strengthening inter-cluster cooperation—especially with naval, rail, ICT, and logistics clusters—will expand opportunities for cross-sector innovation
- The need to incorporate technologies such as artificial intelligence and quantum computing into the sector.

Figure 12 Production subsystem SWOT



Source: Own elaboration based on stakeholders' consultation

2.4. Users and Consumers Subsystem. The multiple value of space for Preparedness

Beyond its economic impact, space technologies and services are essential for national and EU security, resilience and the safety of citizens. Building on a robust physical and digital infrastructure, this section explores the clients and final use of space technologies, space services including Earth Observation, Navigation, Communications, and Space Situational Awareness, to effectively meet the evolving needs of public and private end users across multiple economic sectors A significant share of growth in the global space economy will be driven by downstream space-based data, products and services[61]

More than the 90 % of the Andalusian space production goes to export. The client base of Andalusian space sector companies spans both civil and military domains, reflecting their dual-use capabilities. In the civil sphere, their main markets include international space agencies such as the European Space Agency (ESA) and NASA, commercial satellite operators including Eutelsat, Hispasat and OneWeb, as well as major aerospace primes such as Airbus, Thales Alenia Space, OHB, and Northrop Grumman. As previously mentioned, there isn't any integrator in the region These partnerships are particularly strong in areas such as satellite manufacturing, ground segment systems, solar arrays, precision sensors, and software solutions for navigation, telecommunications and Earth observation.

In the military and security sphere, some space companies in Andalusia maintain long-standing contracts with the Spanish Ministry of Defence, other NATO allied armed forces, providing secure satellite communications, avionics, on-board electronics, and space surveillance systems.

In relation to the sphere of uses and users of space technology relevant to the Andalusian aerospace ecosystem, we will distinguish between the areas of security and defence, those related to public administration, and private users.

2.4.1. Military and Security Users

Following the Communication Vision for the European Space Economy[62], under the current geopolitical context and considering the military capabilities available, defence readiness is impossible without space readiness.

Hence, the development of resilient systems to augment both the safety and security of existing EU space assets and the uninterrupted provision of essential space services across civil sectors, such as transport, energy, and emergency response, will be a critical measure. Enhanced space capacities, crucial for the defence readiness in modern warfare, would therefore comprise:

geo-intelligence data from space with all-weather, very-high resolution imaging capabilities with a high-revisit time.

highly secure and resilient satellite connectivity;

[61] COM(2025)336 final
[62] COM(2025)336 final

secure high-precision navigation positioning and timing services, which are able to withstand jamming and spoofing

The main institutional users of space technologies in Spain are located within the Ministry of Defence and the Armed Forces[63]. The Spanish Air and Space Force operates through the Space Command (MESPA)[64], the Space Surveillance Operations Centre (COVE), and associated assets for situational awareness and operational support. Military deployments abroad, such as those under NATO or EU auspices, are increasingly reliant on secure satellite communications (SATCOM) services, in cooperation with the governmental operator Hisdesat[65] (e.g. SpainSat NG[66]).

While the Air and Space Force leads on surveillance and operational use of space assets, the Army exploits satellite data for intelligence, surveillance and reconnaissance (ISR) and tactical operations; the Navy employs it for maritime awareness and global communications; and the MCCE[67,] ensures the cyber protection of space systems. Together, these elements demonstrate the transversal and multi-domain use of satellite technologies across Spain's Armed Forces.

On the civil side, users include the Guardia Civil and National Police (border security, surveillance of coastal and maritime domains), Civil Protection and the Military Emergencies Unit (UME68) (which activate Copernicus Emergency services), and Maritime Safety Agency (SASEMAR) for maritime domain awareness. At the European level, Spain hosts the EU Satellite Centre (SatCen)69, the Galileo Security Monitoring Centre (GSMC70), and contributes to the EU Space Surveillance and Tracking (EU-SST) network, with the S3T radar located in Morón (Seville).

2.4.2. Civilian, Institutional Users

Space-based technologies are fundamental to strengthening crisis preparedness and response capabilities. The European Union's *Preparedness Union Strategy* highlights that services such as the Copernicus Emergency Management Service and Galileo provide real-time situational awareness, early warnings, and reliable communication when terrestrial networks fail. These capabilities are indispensable for coordinating emergency operations, supporting civil protection, disseminating timely alerts to populations, and mitigating the impacts of both natural and human-induced disasters.

Climate Risks in Andalusia

Andalusia is particularly vulnerable to climate-related risks due to its geographical location, environmental diversity, and intensive exploitation of natural resources. Persistent water scarcity and recurrent droughts threaten urban water supply, agriculture, and ecosystems. Wildfires are increasingly exacerbated by extreme summer temperatures, while coastal erosion and sea-level rise endanger wetlands, beaches, and critical infrastructures. In addition, torrential rainfall events are generating more frequent flash floods, affecting both rural and urban environments.

[63] https://emad.defensa.gob.es
[64] - Actualidad - Asistencia al personal - Detalle pieza información
[65] https://www.hisdesat.es
[66] Programa SPAINSAT NG • Hisdesat
[67] Mando Conjunto del ciberespacio (MCCE) - EMAD
[68] https://www.defensa.gob.es/ume
[69] https://www.satcen.europa.eu
[70] https://www.euspa.europa.eu

These interlinked hazards underscore the need for integrated and evidence-based climate adaptation strategies. In this regard, space-based data and technologies – including Earth observation, satellite communications, and precision navigation systems – provide essential tools for monitoring drought and water use, detecting wildfire outbreaks, assessing coastal change, and forecasting floods. Their application strengthens Andalusia's early warning systems, informs public policy, and enhances the resilience of its economy, environment, and society.

At the regional level, climate adaptation and emergency management involve multiple institutional actors. The Regional Ministry for Sustainability, Environment, and Blue Economy is responsible for guaranteeing access to environmental information in Andalusia and oversees the Environmental Information Network (REDIAM), which integrates and disseminates relevant data. It also leads the Andalusian Climate Action Plan (PAAC 2021–2030).
Meanwhile, the Regional Ministry of the Presidency, Interior, Social Dialogue, and Administrative Simplification, develops and implements policies related to civil protection, emergency response, and wildfire prevention and suppression.

These climate-related challenges, which directly affect public goods, are particularly suited to the launch of Public Procurement of Innovation (CPI) initiatives at the territorial level. One notable example is *Space Innova Andalucía*, a €20 million programme aimed at identifying and applying space-sector innovations across public services. Within this framework, two major challenges have been defined:

Integrated Operational Systems through HAPS Observation

This challenge focuses on developing integrated and strategic operational systems based on High-Altitude Pseudo-Satellite (HAPS) observation. Applications include emergency deployment, as well as prevention and planning in wildfire suppression (under the competencies of the Ministry of the Presidency) and environmental management (under the competencies of the Ministry for Sustainability, Environment, and Blue Economy).

Strengthening the Andalusian Positioning Network

The second challenge is linked to the positioning network managed by the Andalusian Institute of Statistics and Cartography, which falls under the Ministry of Economy, Finance, and European Funds.

The Regional Ministry of University, Research and Innovation leads this initiative, coordinating with several governmental departments. It plays a pivotal role in reinforcing Andalusia's industrial base and enhancing research and development (R&D) capacity within the aerospace sector. The recent conclusions of the preliminary market consultation mark a significant step in the CPI process, paving the way for innovative and strategic solutions tailored to public sector needs.

Space and Agriculture

Space-based technologies are increasingly central to sustainable agriculture, enabling resource efficiency, precision farming, and improved environmental monitoring. Satellite data supports a "less is more" approach by optimising the use of fuel, water, and chemicals, while Earth observation contributes to food security through enhanced crop forecasting and rural development.

In Andalusia, the Regional Ministry of Agriculture, Water and Rural Development has applied satellite-based remote sensing since 2018 for agricultural monitoring, while the Andalusian Institute of Agricultural and Fisheries Research and Training (IFAPA) integrates satellite imagery into its projects.

A milestone was reached in 2023 with the launch of *Platero*, the region's first nanosatellite, in collaboration with LifeWatch ERIC and ceiA3.. *Platero* collects high-resolution images and environmental data, complemented by terrestrial and aquatic sensors via an environmental Internet of Things (e-IoT) network. This nanosatellite delivers real-time diagnostics of agricultural and marine ecosystems, supporting precision farming, fisheries management, and the monitoring of climate change impacts.

2.4.3. Civilian, Non-Institutional Users

While all citizens are ultimately users of space-based services, private companies across multiple sectors are increasingly direct beneficiaries of space-enabled applications. In transport and logistics, satellite navigation and Earth observation underpin fleet management, smart mobility, and the optimisation of global supply chains. In health, telemedicine and satellite communications extend medical services to remote areas and improve emergency response capacity. The construction and infrastructure sectors employ satellite imagery and positioning technologies for surveying, monitoring, and ensuring structural resilience, particularly under changing climatic conditions. More broadly, businesses integrate space-derived data into their operations to increase efficiency, reduce risk, and comply with sustainability standards. The diffusion of these services illustrates how the space industry has become an indispensable enabler of innovation and competitiveness across the private economy.

In Andalusia and Spain, private users and businesses are increasingly dependent on space-enabled applications that extend across transport, health, construction, and infrastructure. In transport and logistics, companies such as Airbus Defence and Space (Seville) and GMV (Madrid, Seville) apply satellite navigation and Earth observation to optimise aviation systems, fleet management, and global supply chains. In maritime transport, the Port of Algeciras integrates satellite data for vessel traffic monitoring and smart logistics. In the health sector, Indra and specialised SMEs provide telemedicine and emergency response solutions based on satellite communications, particularly relevant for rural Andalusia and island territories. The construction and infrastructure sectors employ geospatial data and precise positioning technologies; for instance, Ferrovial and Acciona integrate satellite imagery in large-scale infrastructure projects such as high-speed rail and renewable energy installations.

2.4.4. Expert Insights and SWOT

- Andalusian space companies depend strongly on external markets; exports and international visibility are key drivers.
- The complexity of dual-use and defence technologies exports remains a barrier, despite geopolitical pressure blurring traditional boundaries and future Horizon Europe encompassing dual use technology in the proposal presented by the European Commission.
- There is a need to build a culture of dual technologies and tailored training for businesses.
- Companies face limited knowledge of defence markets, with certification processes and procurement channels representing significant challenges. Creating meeting points with military buyers and leveraging Andalusia's testing infrastructure are seen as opportunities for attracting investment.
- ERDF resources and public procurement of innovation are valued instruments by the business sector, as shown by initiatives like *Espace Innova*; yet, administrations underline the complexity of cost–benefit justification.
- National examples such as Call for QKD (quantum key distribution) projects by CDTI demonstrate the role of public procurement in pioneering new technologies.

- Andalusia must increase its visibility and influence in the European aerospace ecosystem, through higher participation in events and forums and reinforce de organisation of consolidated events in Andalusia such as ADM, International Conference Dual Use and Defence and New Space event.

Figure 13 Users and consumers subsystem. SWOT



Source: Own elaboration based on stakeholders' consultation

3. The desired state of the system.

As stated in the vision of the Andalusian Aerospace Strategy, the desired state of the Autonomous Community aligns with a new regional productive model based on knowledge and innovation. This model grants the aerospace ecosystem systemic and transformative power to drive the broader economy, creating value across multiple dimensions.

In January 2025, when the preliminary results of the expert interviews were subject to consultation, four areas for improvement were highlighted by system stakeholders:

- *Enhancing Internationalisation:* In the current geostrategic context, the space and defence markets are expanding globally, particularly within Europe and NATO. The value chain of the space industry is fully globalised. Strengthening the international presence of Andalusian companies is essential for accessing ESA institutional programmes in which Spain participates and the increasingly significant civil space market. It involves opening doors to new markets, for which Andalusian companies must be well-prepared, given the exceptionally rigorous certification conditions and necessary alliances. It's crucial to leverage the influence of state institutions and have representation in forums to establish alliances and projects. Andalucía TRADE is well-positioned to support this endeavour with its considerable experience.

- *Improving Business Financing Mechanisms*: Private financing for innovation in Andalusia has traditionally been weak, especially in high-investment sectors like space with long-term investments and high technological risk. Although significant efforts have been made by various administrations (EU, national, and regional) to create support mechanisms, they often carry a heavy administrative burden, and their incentive design is not always suitable for small businesses in such a complex sector with specific security and confidentiality requirements. Positive initiatives include the programme launched by the Andalusian Ministry of Industry and Energy, tailored for the space industry and cooperation.

- *Talent Development:* With the growing demand for professionals in a rapidly evolving, high-tech sector that involves international teams and high levels of specialisation, certifications, and processes, developing specialised talent is crucial. Andalusian universities, traditionally focused on the strong aeronautical industry, have begun proposing academic programmes specifically dedicated to the space industry. Similarly, vocational training for specialist technicians has recently gained momentum in Andalusia within the aeronautical sector and could be extended to the space sphere. Moreover, to maintain a competitive space and defence industry, Andalusia needs talent in transformative drivers such as artificial intelligence and quantum technologies.

Given the role space and communication technologies play across other productive sectors, it is essential for professionals in sectors such as agriculture, transportation and logistics, energy, construction, and infrastructure management to have the necessary skills to integrate these technologies. Developing micro-credentials and lifelong learning opportunities for adaptation or retraining is vital to keeping pace with technological changes.

- Tapping into the defence market, leveraging synergies among new space technologies, UAVs, cybersecurity, and integrating domains where Andalusia has strong traditions: naval, land, and air.

Throughout 2025, there has been a global shift reinforcing Europe's need and urgency to bolster its defence capabilities. This urgency has led to an unprecedented number of reports, strategic proposals, new financing instruments, and relaxed budgetary constraints at both the European and national levels. The flow of public—and consequently private—resources into the defence industry is unprecedented, with NATO countries required to annually invest 5% of their GDP in defence-related expenditures. This new situation has reinforced original perspectives on the importance of defence and the potential role of regions like Andalusia in contributing to strategic autonomy and European security. As a result, it has created a window of opportunity for record-level investment in the defence industry.

For this region, the desirable scenario is one that promotes and attracts a maximum number of industrial companies and high-tech space and defence knowledge centres, along with their services. The appropriate measures should create suitable framework conditions for business attraction and the retention and attraction of highly qualified talent.

In this area, there are three dimensions to target, all requiring a commitment to multilevel institutional collaboration. Defence is a state competence, and collaboration between the Junta de Andalucía, the Ministry of Defence, and the Spanish Space Agency is fundamental.

As reiterated throughout the report, the EU and NATO are pressing for a significant increase in defence budgets, projects, and tenders are expanding rapidly. Designing a strategy for exploring and participation in NATO programmes and contracts; supporting initiatives like DIANA to boost defence start-ups; mobilising participation in the European Defence Fund calls; attracting EUDIS events; and enhancing encounters in Andalucía such as ADM or the Dual-use and Defence Technology Conference in collaboration with DG DEFIS and the Network of Defence-related Regions; promoting the involvement of Andalusian actors in EDA activities and ensuring participation in the future European Defence Projects of Common Interest (EDPCI) related to space. In summary, intense efforts to position and showcase the Andalusian ecosystem in the European defence market are necessary, both in its industrial component and its research capabilities and infrastructure.

At the national level, understanding and positioning the region's capabilities in relation to the projects of the Spanish Armed Forces is vital. Activating company participation in the Ministry of Defence's supplier catalogue, understanding procurement mechanisms, necessary certifications, participating in networking activities, and leveraging programmes like COINCIDENTE for incorporating dual-use technologies are critical. An example of this action was the collaboration with DGAM and the industrial office of Future Combat System (FCAS). The collaboration with other regions could improve national coordination and reinforce the national defence technological and industrial base.

Creating collaborative spaces with the armed forces present in Andalusia and incorporating space industry transversally is essential. All army branches rely on technologies grounded in the space industry. Initiatives like the Córdoba Logistic Base exemplify the creation of industrial capacities and technological development tailored to defence needs. Knowledge components, alliances with universities, and infrastructure create a virtuous cycle in Andalusia, which can be magnified by the high presence and territorial distribution of these capabilities, creating new opportunities throughout the region.

An information hub, in collaboration with the Ministry of Defence, the Junta de Andalucía, and enterprises, functioning as a “one-stop shop” for defence market opportunities and procedures, could be pivotal in articulating this opportunity framework.

Preparedness

In recent years, European regions have increasingly faced significant disasters arising from climate change, such as severe floods and unprecedented wildfires, alongside energy-related calamities like widespread blackouts and cybersecurity threats. These events have exposed the vulnerabilities of infrastructure and highlighted the lack of preparedness across the continent. In response, the European Commission launched the Preparedness Strategy 71(European Commission, 2025d) in March 2025, which addresses these challenges and integrates preparedness into the design of EU policies and actions, thereby promoting foresight and anticipation. The strategy aims to enhance the EU’s civilian and military preparedness and readiness for future crises, ensuring that all actors are poised to respond quickly and effectively as needed. It supports national preparedness efforts by improving coordination and efficiency within existing strategies and fostering a culture of resilience through widespread societal engagement.

Key aspects of the Preparedness Union Strategy include improving early warning systems through investment in infrastructure to monitor and detect potential threats across the EU, enhancing Civil-Military Cooperation to ensure effective crisis response, promoting external Partnerships with international entities to address shared risks and enhance collective security, and fostering a preparedness culture that encourages citizen awareness and participation in community-based initiatives. By fostering collaboration among member states, engaging society as a whole and encourage experimentation, this strategy aims to better equip Europe to face future challenges and protect its citizens effectively. Being all that said, it essential to highlight that a major aspect of the Preparedness Union Strategy is that it is positioned under the European competitiveness Compass, as EU has decided to link it to increasing EU competitiveness as preparedness is a driver for transformative innovation.

This new scenario presents an opportunity to incorporate the full dimension of Preparedness into the regional innovation policy planning of Andalusia. This approach, particularly concerning climate change, is already present, in the Andalusian Climate Action Plan (PAAC 2021–2030) and S4 Andalucia and is further exemplified by the public procurement initiative launched by the Andalusian Ministry of University, Research and Innovation, Space Innova, which seeks HAPs-based solutions for firefighting and environmental data collection.

Preparedness means the ability to foresee risks, anticipate, implement measures to prevent them, and respond effectively to crisis when risks or threats materialise. Security, climate, or energy threats can also be understood as "environmental disturbances" or "systemic crises," which can create opportunities for transformative innovation, meaning an innovation that provokes fundamental and systemic changes in economic, social, and environmental systems. Transformative innovation goes beyond incremental improvements by addressing the root causes of societal

[71] Preparedness - European Commission

challenges and fostering inclusive and sustainable development through experimentation, adaptive governance, and coordinated policy efforts (Economic and Social Council, 2025). The transformative scope of innovation would enable new actors to engage with new preparedness challenges, many of which have already been identified by the region, but can be integrated to develop a new vision and coordination process, in which multilevel and interdepartmental collaboration and interterritorial cooperation will drive the design of new initiatives and policy combinations, requiring new roles and, consequently, new capacities and regulatory frameworks.

The European Union's Preparedness Strategy highlights services such as Copernicus, in its Emergency Management Service, and Galileo, which provides real-time situation information, early alerts, and reliable communications when terrestrial networks fail. These capabilities are indispensable for coordinating emergency operations, supporting civil protection, and ensuring timely access to space-based early alert information to directly disseminate alert messages to the population and mitigate the effects of natural and man-made disasters.

For example, the deployment of IRIS² will contribute to the establishment of the European Critical Communications System (EUCCS) to provide critical communications for the missions of public authorities responsible for security and protection throughout the EU and the Schengen area. IRIS² will be crucial in expanding EUCCS coverage in areas with no terrestrial mobile network and improving its resilience against disruptions caused by natural disasters or malicious interferences. In this context, it is essential to enhance cybersecurity and address preparedness and the management of cyber crises across all sectors, including the need for improved communication between administrations within Spain and with other member states and EU entities.

In this regard, the Andalusian Aerospace Strategy already identified a significant opportunity in the downstream segment of the space business. Additionally, the presence of the ESA BIC Sevilla represents an excellent opportunity to facilitate the transfer of space sector technologies, services, and applications to preparedness areas, potentially generating new activities and start-ups in the region.

Regarding support mechanisms, the new scenario has also facilitated the flexibility of fiscal frameworks and the use of European funds. In April 2025, the Communication "A Modernised Cohesion Policy" (European Commission, 2025e) opened the door for territories to adapt their cohesion investments to new strategic priorities, aiming to support stronger, more resilient regions that are better equipped to address geopolitical challenges, including new investment areas such as defence, security, and eastern border regions (European Commission, 2025b). The Commission urges member states and regions to increase support for STEP financing for strategic European sectoral and technological investments and ensure that cohesion-funded investments align with the Union's Preparedness Strategy and the concept of preparedness and security by design when reprogramming under the mid-term review.

Looking forward to the Multiannual Financial Framework (MFF) 2028–2035, the European Commission's proposal72 of 16 July 2025 marks a significant shift in Europe's budgetary priorities,

[72] EU budget 2028-2034

with a fivefold increase in defence and space budgets to €131 billion, representing the largest growth in the entire budget.

The MFF proposal also introduces important changes in the governance of cohesion policy by concentrating the allocation and management of cohesion funds at a national level. Member states and regions will have the possibility to voluntarily support, based on regional needs and priorities, defence-related projects in their national and regional partnership plans. Under this model, regions would need to negotiate with their central governments to ensure the inclusion of funding aimed at regional industrial support measures—such as space and defence projects—in national partnership plans. While this centralisation could improve coordination and help eliminate overlaps in policy areas that are primarily state competencies, it remains to be seen how much latitude regions will retain to design strategies tailored to their specific territorial strengths and innovation ecosystems.

3.1. Orientation and planning subsystem

In the desirable scenario, Andalusia positions itself as a proactive region aligning with European space, security, and defence strategies. Regional and national administrations work in close coordination with AEE, ESA, EUSPA, EDA, and the European Commission - and if possible, with NATO bodies- to simplify access to contracts and to ensure that Andalusian SMEs and start-ups can fully participate. Streamlined procurement and targeted support mechanisms allow local companies to compete effectively, while enhancing innovation capacity and international visibility.

The region strategically leverages its ERDF and ESF resources to promote EU mechanisms such as EDF, PESCO, Horizon Europe, and HEDI to finance scalable, cross-border projects with dual-use potential, ranging from earth observation for agriculture, water and coastal management to secure communications and space-based defence applications. Collaboration between civilian researchers, industry, and defence stakeholders ensures that projects deliver disruptive, versatile services with tangible societal and industrial impacts.

By embedding preparedness and dual-use innovation into governance and planning, Andalusia creates the enabling conditions for regional prosperity: stronger capacity to respond to crises, higher innovation potential, attraction of companies, research centres and investments, increased employment opportunities.

Reinforcing multilevel governance Andalusia would be better positioned to coordinate in the MFF 2028-2035 National and Regional Partnership.

3.2. Resource Mobilisation subsystem

In the desirable scenario, Andalusia develops an agile, predictable, and innovation-friendly framework for financing and human capital that maximises regional prosperity and preparedness. Bureaucratic complexity is significantly reduced by introducing streamlined, digitalised procedures with clear deadlines for applications and payments. Tailored regional programmes address the specificities of the space sector. Public procurement evolves into a more strategic tool, stimulating demand for innovative products and fostering a culture of dual-use development.

At the national level, Andalusia actively aligns with the PERTE Aerospace programme/PTE, positioning its industry in strategic initiatives such as small launchers, Earth Observation constellations, quantum satellite communications, and security-focused EO applications. The region maximises synergies with national financial instruments

At the European level, Andalusian actors engage decisively with ESA, Horizon Europe, EDF, and CSS73 programmes, with simplified procedures that ensure SMEs can compete effectively. Andalusia makes the most of STEP regulation to create synergies with ERDF and gain stronger focus on dual-use opportunities allows the region to build capabilities in critical technologies while gaining international visibility. Cross-border cooperation mechanisms are reinforced, overcoming silo effects between clusters and countries, and ensuring Andalusia is an integral part of large-scale, scalable EU projects.

Private investment is mobilised through improved venture capital availability. Andalusia positions itself as an attractive hub for investors by combining regional support instruments with EU guarantees, enhancing the ability of local start-ups to scale and internationalise.

Human capital development is at the core of the strategy. Andalusian universities introduce specific space curricula, covering mission management, regulations, certification, and dual-use standards. Micro-credential programmes in space technologies, AI, and quantum applications expand the talent base, complemented by lifelong training opportunities for technicians and engineers. Continuous professional education, combined with targeted training for SMEs on defence and dual-use markets, ensures companies are well prepared to access new opportunities.

By embedding preparedness into financing and training strategies, Andalusia creates an enabling environment that strengthens its resilience to crises, attracts investment and enterprises, generates highly qualified employment, and enhances its competitiveness within Europe's strategic space and defence ecosystem.

3.3. Production subsystem

In the desirable scenario, Andalusia develops a robust, integrated, and innovation-oriented production base that allows its space industry to compete effectively at national and European levels while embedding preparedness into its industrial strategy.

The region strengthens the capacity of its SMEs—currently the backbone of the ecosystem—by supporting their integration into European supply chains, facilitating access to subcontracting opportunities, and enabling investment in R&D&I, training, and risk-taking.

To move up the value chain, Andalusia develops or attracts its own integrators—industrial players capable of coordinating complete space missions and large-scale dual-use projects. These integrators consolidate SMEs, research centres, and start-ups into coherent consortia, enhancing the region's role in delivering end-to-end solutions for both European and global markets.

Andalusia Aerospace in collaboration with other clusters, supported by shared facilities provide SMEs with access to specialised infrastructures, testing environments. This approach allows

[73] , "CSS" refers to ESA's "Civil Security from Space"

companies to scale production, retain knowledge, and achieve the critical mass required to respond to EU procurement and industrial programmes.

The Andalusian space industry generates strong synergies with UAVs, ICTs, cybersecurity, and artificial intelligence, and positions itself as a natural partner for major defence companies such as Airbus, INDRA, Navantia, and GDSB. Infrastructures are increasingly opened to the innovation needs of both enterprises and the armed forces, reinforcing dual-use applications and preparedness. The ESA BIC in Seville becomes a key generator of start-ups in the fields of defence, security, and crisis resilience, acting as a pipeline of disruptive innovation for the regional ecosystem.

Cross-border cooperation with Portugal, – without excluding other regional options- is reinforced, creating a structured framework of opportunities for joint development of space infrastructures, satellite constellations, and downstream services relevant to Atlantic and Mediterranean security. Andalusian and Portuguese companies collaborate through clusters, shared facilities, and joint R&D&I projects, positioning the Iberian Peninsula as a strategic European hub for space and defence innovation.

Meanwhile, Andalusian firms evolve into fully internationalised actors, equipped with all the required certifications and the managerial and technological capabilities to lead projects at European and global levels. Their participation in EU strategic initiatives such as the Chips Act, Horizon Europe, EDF and IPCEIS relate to space and defence, consolidates Andalusia as a trusted contributor to Europe's strategic autonomy.

3.4. Users and consumers subsystem

In the desirable scenario, Andalusia positions itself as both a producer and a sophisticated consumer of space technologies, embedding preparedness as a guiding principle across public administrations, armed forces, and private industries. Space services—Earth observation, navigation, secure communications, and space situational awareness—become indispensable tools for strengthening resilience, ensuring security, and driving regional prosperity.

Military and security users rely on Andalusian space companies. Andalusia's ecosystem is recognised as a trusted partner of the Ministry of Defence, Hisdesat, NATO bodies, and allied armed forces. The opening of testing and innovation infrastructures to the needs of enterprises and the armed forces ensures rapid development of dual-use and crisis-response capabilities.

Institutional users—from civil protection to environmental and maritime agencies—embed space-derived data into their daily operations. Andalusia becomes a model region for applying Copernicus, Galileo, to climate resilience: monitoring droughts, wildfires, floods, and coastal change in real time; strengthening early warning systems; and informing public policies through integrated digital platforms such as REDIAM. Public procurement of innovation (e.g. Space Innova Andalucía) consolidates the region as a testbed for space-enabled solutions that directly improve public services and territorial resilience.

Private users across transport, health, energy, construction, and infrastructure increasingly adopt space applications. Andalusian ports, logistics firms, and aviation companies leverage satellite navigation and Earth observation for smart mobility and supply chain optimisation; construction and renewable energy leaders integrate geospatial data for resilient infrastructure; and telemedicine expands healthcare coverage through satellite communications. These downstream applications illustrate how space becomes a transversal enabler of competitiveness, sustainability, and crisis preparedness in the private economy.

The ESA BIC in Seville acts as a generator of start-ups specialising in defence, security, and preparedness, while Andalusian firms—fully internationalised, with the necessary certifications—evolve into trusted partners capable of leading European and global projects. Synergies with UAVs, ICTs, AI, and cybersecurity strengthen dual-use applications, and large defence companies (Airbus, Navantia, GDSB) become strategic anchors for innovation ecosystems.

Cross-border cooperation with Portugal creates a structured framework for shared infrastructures, satellite constellations, and downstream services relevant to Atlantic and Mediterranean security, positioning the Iberian Peninsula as a pan-European hub of space and defence users. Membership in NEREUS and active participation in the ECCP network connect Andalusia to a European community of space regions.

By embedding preparedness into the user and consumer dimension—ensuring demand-driven innovation, dual-use adoption, and cross-sector integration—Andalusia consolidates an ecosystem where space is not only a driver of exports but also a cornerstone of regional resilience, prosperity, and strategic autonomy.

A summary of the desired state of the system is presented in Figure 14.

Figure 14. Desired state of the Andalusian Ecosystem

Orientation and Planning Subsystem

- Andalusia positions itself as a proactive region aligned with European space, security, and defence strategies.
- Close coordination between regional and national administrations and European bodies (AEE, ESA, EUSPA, EDA, the European Commission, and ideally NATO).
- Simplified access to contracts and targeted support mechanisms enabling Andalusian SMEs and start-ups to participate effectively.
- Strategic use of ERDF and ESF resources to promote participation in EDF, PESCO, Horizon Europe, and HEDI.
- Promotion of scalable, cross-border dual-use projects in areas such as Earth observation, water and coastal management, secure communications, and space-based defence applications.
- Strong collaboration between civil research, industry, and defence stakeholders to deliver disruptive services with tangible societal and industrial impact.
- Embedding preparedness and dual-use innovation into governance to foster resilience, innovation, employment, and investment.
- Reinforced multi-level governance to strengthen Andalusia's capacity to coordinate within the 2028–2035 Multiannual Financial Framework.

Resource Mobilisation Subsystem

- Development of an agile, predictable, and innovation-friendly financing.
- Creation of regional programmes tailored to the specific needs of the space sector and the use of public procurement as a strategic tool for innovation and dual-use development.
- Strong engagement in European programmes (ESA, Horizon Europe, EDF, CSS) with simplified procedures to ensure SMEs' participation.
- Effective use of the STEP Regulation to build synergies with ERDF and focus on dual-use opportunities.
- Mobilisation of private investment through enhanced venture capital availability and EU-backed guarantees.
- Development of human capital through specialised university curricula in space mission management, regulation, and certification; micro-credentials in AI and quantum technologies; and continuous training for technicians and SMEs on defence and dual-use markets.
- Integration of preparedness into financing and training strategies
- Membership in NEREUS and participation in the ECCP network connect Andalusia to a wider European community of space regions.

Production Subsystem

- Development of a robust, integrated, and innovation-oriented industrial base capable of competing nationally and across Europe.
- Strengthening of SMEs by facilitating their integration into European supply chains, subcontracting opportunities, and access to R&D&I investment and training.
- Creation or attraction of regional integrators able to coordinate full space missions and large-scale dual-use projects.
- Collaboration among clusters such as Andalusia Aerospace and access to shared facilities and test environments to enable industrial scaling.
- Generation of synergies with UAVs, ICT, cybersecurity, and artificial intelligence sectors.
- The ESA BIC in Seville serves as a generator of start-ups in defence, security, and crisis resilience, feeding innovation into the regional ecosystem.
- Reinforced cross-border cooperation with Portugal, enabling joint development of infrastructures, satellite constellations, and downstream services relevant to Atlantic and Mediterranean security.
- Andalusian companies evolve into fully internationalised actors, with the certifications and capabilities required to lead European and global projects.
- Active participation in strategic EU initiatives such as the Chips Act, Horizon Europe, EDF, and IPCEIs, consolidating Andalusia as a trusted contributor to Europe's strategic autonomy.

Users and Consumers Subsystem

- Andalusia positions itself as both a producer and sophisticated consumer of space technologies, embedding preparedness and resilience across public and private sectors.
- The regional ecosystem becomes a trusted partner of the Ministry of Defence, Hisdesat, NATO bodies, and allied armed forces.
- Institutional users (civil protection, environmental, and maritime agencies) systematically integrate Copernicus and Galileo data into operations for climate resilience, real-time monitoring
- Public procurement of innovation initiatives (e.g. Space Innova Andalucía) make the region a testbed for space-enabled solutions improving public services and territorial resilience.
- Private users in transport, logistics, energy, construction, and healthcare increasingly adopt space applications for smart mobility, supply-chain optimisation, resilient infrastructure, and telemedicine.
- Synergies with UAVs, ICT, AI, and cybersecurity reinforce dual-use innovation, with major defence companies (Airbus, INDRA, Navantia, GDSB) acting as anchors for the regional ecosystem.
- Cross-border collaboration with Portugal establishes shared infrastructures and satellite services, positioning the Iberian Peninsula as a European hub for space and defence innovation.
- By embedding preparedness into user and consumer dimensions, Andalusia builds a demand-driven, cross-sector innovation ecosystem where space becomes both an export driver and a cornerstone of regional resilience and prosperity.

Source: Own elaboration

4. Conclusions, recommendations and next steps

4.1. Conclusions and Recommendations

The following set of recommendations, drawn from stakeholder consultations and expert interviews, aims to contribute to the realisation of this long-term vision.

1. Scale up Andalusia's Space Industry through Defence Integration and International Expansion

Andalusia space sector is a growing domain which can grow in the national and international market exploring both in civil and markets. This recommendation proposes supporting both civil and defence-oriented activities, business models, and investment strategies in the space sector and related technologies to take advantage of the historic window of opportunity for investment in defence and space to attract project, companies and talent.

It is proposed to encourage the integration of Andalusian actors (SMEs, research and innovation centres) to European and national value chains through integrators, business cooperation and interterritorial cooperation with other and alliances with prime contractors and strengthen multilevel coordination and governance (Regional Government–Ministry–Spanish Space Agency–EU/NATO)

Improved access to AEE, MdD, ESA, European Union and NATO programmes and boosting internationalisation.

- Permanent liaison with the Spanish Space Agency (AEE) and Ministry of Defence: Establishment of an Andalusian focal point to ensure effective participation in national programmes and improved visibility of regional capabilities in calls for proposals.
- Design an initiative for enhancing internationalisation, exploring and promoting the participation of Andalusian entities in ESA, DG DEFIS, Horizon Europe, Digital Europe, EDA and NATO programmes and contracts, in collaboration with AEE and Ministry of Defence to increase the returns of these programmes in Andalusia, making the most of the Andalusian node of the Enterprise Europe Network - CESEAND
- Increase Andalusia's representation in space and defence international forums, to gain visibility, facilitate the entrance of Andalusian entities and, when possible, influence programme design (TEDAE, PAE..)

- Facilitate Andalusian participation in EDF, EUDIS, Cassini, and DIANA with targeted calls for SMEs and start-ups.
- Reinforce encounters such as ADM and Dual-use and Defence Technology Conference, in collaboration with DG DEFIS and the Network of Defence-related Regions.
- Support cross-border projects with other European regions, especially with Portugal (Atlantic constellations, defence innovation, maritime surveillance).
- Collaboration with NEREUS, European Cluster Collaboration Platform and other international networks to facilitate knowledge sharing, collaboration, and best practices among member regions,

thereby enhancing Andalusia's capacity to engage in space initiatives and leverage opportunities at the European level.

Foster defence integration and dual-use culture

- Reinforce partnerships with defence primes (INDRA, Airbus, Navantia, GDSB, Rheinmetall, Escribano, SENER, Thales Alenia, OHB) to anchor Andalusian SMEs in European supply chains.
- Promote dual-use demonstration projects (satcom, ISR, UAV integration, cyber resilience) supported by regional, national, and EU funds.
- Establish regular meeting points between Andalusian SMEs and defence buyers (Ministry of Defence (DGAM), NATO, EDA).
- Map regional capabilities in relation to the Spanish Armed Forces' projects.
- Encourage closer engagement between the Air and Space Force and SMEs, including initiatives such as SME Demo Days
- Strengthen participation in COINCIDENTE and other Ministry of Defence innovation programmes.
- Update the MoU JA – DGAM, involving the new Direction of Strategy and Innovation and reinforce coordination with the MINISDEF- Armed forces to improve the regional awareness of the contracts and programmes where the Andalusian companies can participate.
- Enhance collaboration with other Spanish regions to promote business cooperation and reinforce to the national defence technological and industrial base (DTIB).
- Establish an observatory for dual-use technologies and defence needs in collaboration with the Ministry of Defence

Reduce Access Barriers for Start-ups and SMEs in Defence Projects

- Implement initiatives to lower entry barriers for start-ups and SMEs in regional, national, and transnational defence projects, facilitating their integration into the market. (mentorship, training, networking events…)
- Provide advanced services and financial support for certification, audits and consultancy services to SMEs transitioning into the defence sector, providing support to companies in achieving required certifications and navigating procurement mechanisms.
- For the European Commission, clarify and simplify regulations related to dual-use technologies, ensuring SMEs and research institutions understand their obligations under EU Regulation 2021/821 (Dual-Use Regulation)

Support dual use test beds and demonstration infrastructures

- Maximise the impact of research and test infrastructures (CATEC-ATLAS, CEUS/CEDEA, IMSE-CNM, CNA, UMA Laser Lab) by connecting them with industrial consortia and defence users.
- Upgrade technological infrastructures and innovation centres to attract new industrial investment to the regions and improve industrial competitiveness and to support defence and civil test and certifications.

2. Streamline access to multiple funding sources for strategic projects and enhance incentive programmes

The fragmentation of funding regulations across European, national, and regional operational programmes poses a significant challenge to the effective coordination and synchronisation of resources. This is further compounded by administrative rigidity and a risk-averse culture within public administrations, where concerns over non-compliance or audit sanctions discourage the support of high-risk, high-reward initiatives. As a result, both public and private research actors encounter substantial barriers to pursuing ambitious innovation agendas, particularly those requiring cross-sectoral cooperation and ecosystem development. The ongoing flexibilization of the Cohesion Policy framework following the midterm review—alongside the anticipated implementation of the Defence Readiness Omnibus—offers a timely opportunity to facilitate integrated funding approaches.

Making the most of the possibilities of the Strategic Technologies for Europe Platform Regulation (STEP)

- Leverage STEP as a basis for designing new incentive mechanisms and fostering synergies between regional cohesion policy instruments and European strategic programmes such as the European Defence Fund, Horizon Europe, the Digital Europe Programme, the Innovation Fund and related space projects of common interest. The STEP Seal is intended to promote top-rated proposals for access to cumulative or combined funding and to attract additional or alternative private and public finance.
- Co-financing schemes: Development of national–regional co-investment mechanisms to support regional (Andalusian) SMEs and mid-caps in ESA, Horizon Europe, and EDF or EDIP projects, particularly those requiring scale and critical mass.
- Reform national and EU-level financing rules to better support dual-use infrastructure and innovation projects, enabling greater alignment with long-term strategic priorities and technological sovereignty goals.

Reinforce synergies between ESA, EUSPA national and regional programmes

- Integrate ESA–EUSPA synergies into national funding instruments (such as the Aerospace PERTE and the National Space Technology Plan – PTE), enabling Spanish companies to develop components, software, and services that comply with ESA frameworks while targeting EUSPA market uptake calls.

Design effective incentive programmes

- Simplify and expedite administrative processes and requirements for small companies, reducing bureaucracy while ensuring deadlines for procedures and payments are consistently met.
- Centralise sector-specific information through the creation of a regional observatory to improve coordination and transparency.
- Design dual-use incentive programmes that accommodate the necessary conditions of confidenciality and security required by the sector.
- Develop dedicated regional funding programmes for the space sector to better address its high technological risks and unique requirements.

- Adjust financial criteria imposed on SMEs, such as accounting standards and penalties for late payments, to better align with the realities of deep-tech companies. Consider alternative criteria, including employee salary levels and technological sophistication.
- Design dedicated financing instruments to challenge-driven initiatives propose by primes or public Administration, targeting SMEs, start-ups, and challenge-driven initiatives promoting open innovation and experimentation.

3. Stronger Industry ready to compete in the global market

Scaling up and strengthening Andalusian Companies

- Launch a regional scale-up programme specifically designed for space SMEs, combining financial incentives, equity funds, and mentoring to enhance their size, investment capacity, and innovation potential.
- Support the establishment or attraction of Andalusian space integrators that can coordinate comprehensive missions and dual-use projects, ensuring SMEs access higher segments of the value chain.
- Attract investment of industry and research centres which complement Andalusian capacities, following the successful installation of the IMEC to Malaga.
- Promote business cooperation to build critical mass and competitiveness at both regional and national levels, through local calls and programmes like PTE and CDTI.
- Design regional funding programmes for companies and research institutions to strengthen technological and industrial capacities, enabling them to qualify for national and international space and defence projects. This builds on the recent call launched by the Regional Ministry of Industry and Energy to support industry participation in the life cycle of new satellite systems.

Empower and strengthen clusters related to strategic technologies.

Reinforce the role of clusters, such as Andalucia Aerospace and promote cross-cluster cooperation (aerospace, naval, rail, ICT, cybersecurity, UAVs) to foster critical mass and convergence of technologies to foster dual-use innovation and cross-sectoral synergies. Having an active presence in the European Network of Aerospace Clusters (EACP)

Reinforce regional capacities in Artificial Intelligence and Quantum Technologies—across both private industry and academia—to keep pace with technological advances and safeguard regional competitiveness

4. Whole-of-government coordination, overcoming policy fragmentation

This includes engagement with key institutions such as the Spanish Space Agency, Ministry of Defence and improved management of boundaries across government departments and policy areas. A whole-of-government approach is essential for navigating the complex interdependencies

across sectors that shape security and defence policy—such as education, innovation, digitalisation, industrial strategy, and regional development.

European level

- Creation of a coordination Board participated by AEE-ESA, INTA and Andalusia Regional government
- Strengthen Andalusia's awareness of space and defence policies, market developments, and industry trends. As an example, position papers analysing the impact of European political decisions on the Andalusian ecosystem could provide valuable input for regional policymaking.
- Enhance the role of the Andalusian Government's Brussels delegation and formalise regional representation in EU networks such as ASD, ERRIN, EURADA, ENDR, and NEREUS, ensuring that Andalusian priorities are effectively channelled into EU strategic agendas.

National level

- Reinforce Andalusian representation in national networks such as TEDAE74 (for industry), Spanish Aerospace Platform75 (industry and academy) ensuring regional priorities are channelled into National and EU strategic agendas.
- Integration into national defence technology roadmaps: Ensuring Andalusian research and industrial strengths (e.g. UAVs, microelectronics, advanced manufacturing) are incorporated into Spain's space and defence strategies.

Regional level (Andalusia)

- Whole-of-government coordination body: Create a Space-Defence Coordination Task Force within the governance of the Aerospace Strategy, involving the Regional Ministries of Economy, Industry, Universities, Innovation, Education, Employment, Environment, Agriculture, Education, etc., as well as companies, universities and technology centres.

At the local level

- Municipal innovation districts: Empowering Seville, Málaga, Cádiz, and Huelva as hubs hosting specific infrastructures (e.g. Sevilla Aerospace District, Málaga TechPark, Cádiz manufacturing technologies)

5. Foster Market creation through demand-driven instruments

[74] Qué es Tedae - TEDAE
[75] PAE – Plataforma Tecnológica Aeroespacial Española – Plataforma Tecnológica Aeroespacial Española

The interviews and stakeholders' consultation highlighted that a key measure to strengthen Andalusia's space and dual-use ecosystem is the promotion of innovation-oriented public procurement (PPI). International experience shows that when governments act as first customers for early-stage space solutions (e.g. NewSpace platforms, HAPS), they provide critical market validation and trigger private investment. The experience of the SPACE INNOVA public procurement of innovation (CPI) programme, launched by the Regional Ministry of University, Research and Innovation, together with CPI initiatives promoted by CDTI, is highly valued by companies as a means of generating opportunities and gaining early market validation.

However, representatives of procuring authorities have expressed concerns about the complexity of these procedures and the excessively long timeframes often required for their implementation.

- Challenge-driven procurement frameworks—where the public sector formulates needs for which no solutions currently exist—can orient innovation towards strategic priorities, encouraging start-ups to invest in R&D with clear market opportunities.
- Capacity-building for public procurers, including training on innovation procurement, legal frameworks, and technology assessment to expand innovation-oriented public procurement to reduce adoption risks.
- EU institutions, nation and regions may consider revising procurement practices to move beyond lowest-price criteria by prioritising performance, innovation, security and strategic value.
- Multilevel and cross border collaboration and best practices exchange in pre-commercial procurement (PCP) and pilot schemes focused on space.

6. Address the talent and skills gap in strategic technologies

A systemic approach to talent development that bridges universities, vocational training, and industry needs, while fostering continuous learning in emerging technologies is required. In this context, the following measures are proposed to strengthen Andalusia's human capital base and ensure that SMEs and startups can compete effectively.

- Strengthen space-specific curricula in Andalusian universities (degrees, masters, micro credentials) with dual-use orientation. The flamant MOSE of Sevilla University and the micro credentials launched are excellent examples of the new talent support instrument.
- Establish lifelong training programmes for technicians and engineers in emerging technologies (AI, quantum, cybersecurity, robotics).
- Develop joint training programmes between companies, universities, research centres, and the armed forces to close gaps in knowledge transfer.
- Training programmes for public officials, researchers, and industry—developed in partnership with European institutions such as the European Security and Defence College[76] and Andalusian Universities —strengthen skills in space-based services for security, resilience, and risk management, all areas highly relevant for Andalusia.

[76] https://www.esdc.europa.eu/index_en

- Promote the participation of the Andalusian Universities in the Doctorate School of the European Security and Defence College.

7. Boost Preparedness as a driver of Transformative Innovation

The European Preparedness Strategy inspires the integration of a new dimension of preparedness by design into the regional innovation policy planning of Andalusia, S4 Andalusia. Issues such as security, climate, and energy can be seen as "landscape shocks"—disruptive events that also create windows of opportunity for transformative innovation, understood as innovation that brings about fundamental and systemic changes in economic, social, and environmental systems.

To incorporate anticipatory thinking (Science et al., 2024), it is essential to develop foresight processes to assess the main risks and threats faced by Andalusia. By adopting exploratory tools[77] and participative methods, policymakers can explore risks and interconnections, thereby gaining a systemic understanding.

As a result of these experimentation process, policymakers can identify policy interventions. By dissecting the root causes of crises, decision-makers can brainstorm potential policy responses, identifying leverage points and partnerships aimed at mitigating negative impacts. While the following mission-oriented initiatives serve as illustrative examples, actual missions would emerge from the comprehensive foresight and policymaking decisions:

M1. Andalucía Zero Avoidable Fire

A mission to prevent and mitigate wildfires through comprehensive early detection, rapid autonomous initial attack, and socio-ecological resilience. It integrates Copernicus Emergency Management Services, high-altitude platforms, drones, and artificial intelligence analytics, thereby enhancing the effectiveness of civil protection systems and fostering multilevel governance and citizen involvement to safeguard rural territories.

M2. Floods and Droughts Under Control

This mission seeks to address hydrological risks by deploying digital twins of river basins, IoT-based sensors, and advanced forecasting models. Linked with Copernicus EMS and ECMWF data services, it provides actionable information to civil protection units, early warnings, citizen guidelines, exercises, and simulations, all within a multilevel governance framework that encourages citizen involvement.

M3. Regional Cyber-Resilience SOC (NIS2-ready)

[77] Polycrisis exploration workshop | Knowledge for policy

The creation of a regional Security Operations Centre to provide real-time monitoring, threat detection, and incident response for public administrations, operators of essential services, and SMEs in critical sectors. Fully aligned with the EU NIS2 Directive, it acts as a hub for cyber-resilience, offering early-warning capabilities and ensuring continuity of services through coordinated response across Andalusia, Spain, and the EU.

M4. Preparedness for Armed Conflicts at European Borders

This mission enhances Andalusia's readiness to face the ripple effects of armed conflicts on Europe's borders, where hybrid threats increasingly combine military, cyber, and energy disruptions. It promotes a whole-of-government and whole-of-society approach, engaging public authorities, defence agencies, industry, academia, and civil society in joint preparedness efforts. Andalusia's established strengths in aerospace, naval, land, and unmanned systems enable dual-use innovation and integration into European defence and security value chains.

Space-based services provide indispensable support: Copernicus for situational awareness and border monitoring, Galileo for secure navigation and emergency warning, and the future IRIS^2 for resilient cross-border communications. These assets underpin civil-military coordination, protect critical infrastructures, and strengthen Andalusia's contribution to the European Preparedness Union.

4.2. Next Steps

This report provides a timely foundation for Andalusia's next steps at the European level. The Regional Government's forthcoming collaboration in the Experimentation Journey on Territorial Preparedness, under the Preparatory Action "Innovation for Place-Based Transformation" led by the Joint Research Centre and financed by the European Parliament, represents a unique opportunity to translate the insights of this report into concrete pilot action in collaboration with other European Regions. By aligning regional innovation strategies with European foresight and preparedness agendas, Andalusia can reinforce its role as a frontrunner in dual-use innovation, strengthen multilevel governance capacities, and contribute actively to the shaping of Europe's collective resilience.

Figure 15. Summary of Final Recommendations

1. Scale up Andalusia's Space Industry through Defence Integration and International Expansion

- Improved access to Spanish Space Agency, Ministry of Defence, European Union, ESA and NATO programmes and boosting internationalisation
- Foster defence integration and dual-use culture
- Reduce access barriers for start-ups and SMEs in Defence Projects (EUDIS-EDF)
- Support dual use test beds and demonstration infrastructures

2. Streamline access to multiple funding sources for strategic projects and enhance incentive programmes

- Design effective Incentive Programmes for the Space Sector
  Make the most of the ERDF-Andalusia's synergies with European initiatives (HE, EDF, Connecting Europe,
- etc.) through the STEP seal
- Reinforce synergies between ESA, EUSPA national and regional programmes

3. Stronger Industry ready to compete in the global market

- Scale up and strengthening Andalusian companies
- Empower clusters related to strategic technologies
- Reinforce regional capacities in artificial Intelligence and quantum technologies

4. Whole-of-government coordination, overcoming policy fragmentation

- Creation of a coordination board participated by AEE-ESA, INTA and Andalusia Regional government
- At regional level, creation of a Space-Defence Coordination Task Force within the governance of the Andalusian Aerospace Strategy

5. Foster Market creation through demand-driven instruments

- Challenge-driven procurement frameworks—where the public sector formulates needs for which no solutions currently exist—can orient innovation towards strategic priorities, encouraging start-ups to invest in R&D with clear market opportunities
- Capacity-building for public procurers
- Public Administration may consider revising procurement practices to move beyond lowest-price criteria by prioritising performance, innovation, security and strategic value
- Multilevel and cross border collaboration and best practices exchange in pre-commercial procurement (PCP) and pilot schemes focused on space

6. Address the talent and skills gap in strategic technologies

- Strengthen space-specific curricula in Andalusian universities (degrees, masters, micro credentials) with dual-use orientation
- Establish lifelong training programmes for technicians and engineers in emerging technologies (AI, quantum, cybersecurity, robotics)
- Develop joint training programmes between companies, universities, research centres, and the armed forces.
- Training programmes for public officials, researchers, and industry—developed in partnership with European institutions such as the European Security and Defence College
- Promote the participation of the Andalusian Universities in the Doctorate School of the European Security and Defence College

7. Boost Territorial Preparedness as a driver of Transformative Innovation

- Develop foresight processes to assess the main risks and threats faced by Andalusia, to explore risks and interconnections, thereby gaining a systemic understanding
- Participate in the Experimentation Journey on Territorial Preparedness in collaboration with other European Regions and supported by JRC and the European Parliament

Source: Own elaboration

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List of abbreviations and definitions

Abbreviations	Definitions
AEE	Spanish Space Agency (Agencia Espacial Española
AM	Additive Manufacturing
CARD	Competitive Space Start-ups for Innovation Initiative (EU space entrepreneurship support)
COPERNICUS EMS	Copernicus Emergency Management Service
COINCIDENTE	Cooperation in Scientific Research and Development in Strategic Technologies (Spain)
CSDP	Common Security and Defence Policy
DG DEFIS	Directorate-General for Defence Industry and Space (European Commission)
DIANA	Defence Innovation Accelerator for the North Atlantic (NATO)
EDA	European Defence Agency
EDF	European Defence Fund
EDIP	European Defence Industry Programme
EDIS	European Defence Industrial Strategy
EGNOS	EGNOS – European Geostationary Navigation Overlay Service

Abbreviations	Definitions
ESETA	European Statistics on the Space Economy (initiative)
ESETA	ESETA – European Statistics on the Space Economy
EUDIS	EU Defence Innovation Scheme
EUSPA	EU Agency for the Space Programme Centre
GNSS	Global Navigation Satellite System
GOVSATCOM	Governmental Satellite Communications (EU programme)
GSMC	Galileo Security Monitoring
HAPS	HAPS – High-Altitude Pseudo-Satellite
IRIS²	Infrastructure for Resilience, Interconnection and Security by Satellite
ISR	Intelligence, Surveillance and Reconnaissance
NDIS2 / NIS2 –	Directive on Security of Network and Information Systems 2

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Annex 1. Stakeholders' consultation

Agenda – Stakeholders Consultation Workshop

Date: 24 January 2025

Venue: Regional Ministry of Industry and Energy, Junta de Andalucía Headquarters, Calle Castelar 22, Seville

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09:30 – 10:00 | Institutional Welcome

Cristóbal Sánchez, Deputy Minister of Industry and Energy, Junta de Andalucía

Mikel Landabaso, Director, Joint Research Centre – Seville

10:00 – 10:20 | Spanish Space Technological Plan

José María Pérez, Head of National Programmes Department, Spanish Space Agency

10:20 – 10:35 | Support Instruments of the Regional Ministry of Industry, Energy and Mines

José Javier Alonso Membrives, Secretary-General for Industry and Mines

Elena María Ruiz Agudo, Head of Incentives Service

10:35 – 10:45 | Presentation of the REGDUALOSA Research Action (Regions–Dual Use–Open Strategic Autonomy): The Andalusia Case Study

Carmen Sillero, Joint Research Centre

Esperanza Caro, Regional Ministry of Industry and Energy, Junta de Andalucía

10:45 – 11:00 | The Socio-Technical System of Space Technology in Europe

Raúl González, University of Leicester (online)

11:00 – 11:15 | ESETA: Creation of the First Official European Statistics on the Space Economy

Luis Galiano, Joint Research Centre

11:15 – 11:30 | Analysis of the Complexity of the Space Industry in Andalusia

Carla Sciarra, Joint Research Centre

11:30 – 12:00 | Coffee Break

12:00 – 14:00 | Stakeholder Consultation on Preliminary Results of the Andalusia REGDUALOSA Case Study on Space

Participation of Andalusian space ecosystem stakeholders

Moderated by: Melanie Durth, REGDUALOSA Expert

14:00 | Conclusions and Closing

Regional Ministry of Industry and Energy, Junta de Andalucía

Annex 2. Interviews to the experts

Experts interviewed

ENTITY	NAME	POSITION
INDUSTRY		
SOLARMENS	José Migel Moreno	CEO
ALTER TECHNOLOGIES	Eva Díaz	Business Development Manager
FOSSA SYSTEM	Vicente González	CTO
INTEGRASYS	Pedro Ruiz	R&I Manager
OPEN COSMOS	Aurelio Marti	Business Development Director
AYTANA	Miguel Angel Jiménez Roldán	HO Aerospace and Defence Division
SAFRAN N&T	Pablo Marin	R&T Business Development
CATEC	Jesús Boby	Smart Industry Director
ESCRIBANO M&E	Gabriel Coloma	Business Development Director for EU and NATO
NOVAINDEF	Fernando Lasangni	CTO
Grupo OESIA	Salvador Alvarez Pascual	Deputy Director for Strategy
UNIVERSITIES		
University of Seville	Rafael Vázquez Valenzuela	Professor
University of Cádiz	Ignacio Mateos	Professor
PUBLIC ADMINISTRATION		
Regional Ministry of Industry and Energy	Elena Ruiz Agudo	Head of Service for Industrial Promotion
Regional Ministry of Economy, Treasury and European Funds	Juana García Simón	Head of Service for European Funds
Regional Ministry of Economy, Treasury and European Funds	Patricia Eguilior	Technical Counsellor
Regional Ministry of University, Research and Innovation	Manuel Ortigosa Brun	Technical Counsellor
FINAL USERS		
DIH AGROTECG	Ana Belén Cabezas	Deputy Coordinator

Regional Ministry of Agriculture, Fisheries, Water and Rural Development	Maria del Mar Cátedra Cerón	Technical Counsellor

Source: Own elaboration

The interviews were conducted individually via Teams and lasted between 30 minutes and 1 hour, depending on the experts' availability. Additionally, any email or phone exchanges necessary to clarify any points raised during the interview were conducted.

Questionnaire for Industry and Technology Centres

1. Industrial Relevance

- Turnover (2023)
- Employment (2023)
- Exports (% of total and countries)
- Participation in international fairs (2024/2025)? Please specify.
- Has your organisation received public support for internationalisation? From which authority (regional/national)?
- Networks: key entities engaged at regional, national, European, and international levels.

2. Financing

- Total funding received in 2023.
- Sources of funding:
    - Private (% and specify financial entity)
    - Public (% and specify programme/agency)
- Innovation funding: Number of projects and calls (regional, national, European). Were you a leader or subcontractor?
- Comments to improve public financing schemes.
- Main challenges encountered in accessing public/private financing.

3. Skills and Training

- Origin of employees (% from):
    - Universities
    - Vocational training
    - Other sources
- Training needs identified.
- Suggestions for future programmes.

4. Industrial Policies & Dual-Use Strategy

- Which policies most affect your performance? (taxation, innovation, industrial, environmental—indicate authority: regional, national, European).
- Which funding programmes have the most impact on your business (regional/national/European; civil/defence)?
- Do you see a need for greater coordination between civil space and defence policy? How could this synergy be improved?
- Recommendations for the Regional Government (Junta de Andalucía) to complement EDF initiatives.

5. End Users

- What type of end users adopts your technology? Public or private?
- Civil or defence sector?
- Where are they located? Regional, national, European, international?

Questionnaire for Universities

1. International Impact

- Participation in international fairs/conferences (2024/2025)? Please specify.
- Has your institution received public support for internationalisation? From which authority (regional/national)?
- Networks: key entities engaged at regional, national, European, and international levels.

2. Financing

- Total funding received in 2023.
- Sources of funding: Regional, national, or European (specify programme/fund).
- Comments to improve public financing schemes.

3. Education and Training

- Current programmes in space-related disciplines:
    - Undergraduate degrees
    - Master's degrees
- Student demand (high/low).
- Graduate employment opportunities: public or private sector.
- Forecast of future programmes.

4. Industrial Policies & Dual-Use Strategy

- Which policies have the greatest impact on your development (regional/national/European; civil/defence)? Please specify authority.
- Do you see a need for greater coordination between civil space and defence policy? How could this synergy be improved?
- Suggestions to improve regional, national, or European policies.

5. End Users

- Collaboration with industry?
- What type of end users utilise your research/technology (public/private, civil/defence)?
- Location of end users: regional, national, European, international.

Questionnaire for Public Administrations

1. Strategy

- Which strategies currently include the space sector?

2. Financing

- What types of funding are provided to the space industry?
- Are all available funds fully utilised?

3. Innovation

- What types of funding mechanisms support innovation in the space industry?
- Are these funds fully utilised?

4. Skills and Training

- What support/programmes exist to promote training for the space sector?
    - University
    - Vocational training
    - Other
- Forecast of future programmes.

5. Industrial Policies & Dual-Use Strategy

- What actions are being implemented to foster dual-use policies in the space sector?
- How could synergies between civil and defence space policies be improved?

Questionnaire for End Users

1. Technological Solutions

- Do you find the technological solutions you need within Andalusia?
- What additional capabilities should be developed?

2. Industrial Policies & Dual-Use Strategy

- Which policies have the most impact on your business (regional/national/European; civil/defence)? Please specify authority.
- Do you see a need for greater coordination between civil space and defence policy? How could this synergy be improved?

Annex 3. List of Andalusian space companies

List of the main space companies identified in the region:

NEURODIGITAL TECHNOLOGIES, S.L.	AIRBUS DEFENDE AND SPACE	GRUPO UMI AERONÁUTICA (MEUPE)
ROBONITY: WORLDWIDE TECH STARTUP	AIRTIFICIAL AEROSPACE & DEFENSE, S.A.	INDRA-INDRA SISTEMAS, S.A.
GTD, INGENIERIA DE SIWSTEMAS Y DE SOFTWARE, S.A.	ALTER TECHNOLOGY TÜV NORD S.A.U	INGENIERIA Y SOLUCIONES INFORMÁTICAS DEL SUR, S.L., ISOIN
TAITUS SOFTWARE	ALTRAN	INTEGRASYS, S.A.
BLACK HAT, S.L.	ANAFOCUS	LATESYS IBERIA
SEVEN SOLUTIONS	ANTENNA SYSTEM SOLUTIONS	OESIA GRUPO
TRIANA SCIENCE & TECHNOLOGY	ARQUIMEA INGENIERÍA	OPEN COSMOS
SOL GALAXY	BRAMAH SOLUCIONES, S.L.	OPTIMIZACIÓN ORIENTADA A LA SOSTENIBILIDAD, S.L. (IDENER)
ZERO 2 INFINITY, S.L. (Z2I)	CELESTIA TECHNOLOGIES GROUP	SAFRAN E&D
DHV TECNOLOGÍA ESPACIAL AVANZADA MALAGUEÑA, S.L.	CESA	SIEMENS
MÁLAGA AEROSPACE, DEFENSE AND ELECTRONICS SYSTEMS, S.A. (MADES)	CT INGENIEROS	SKYLIFE ENGINEERING, S.L. ES

MLABS OPTRONICS	ELEMENT MATERIALS TECHNOLOGY SEVILLE	SOLAR MEMS TECHNOLOGY, S.L.
NETWORK & SATELLITE SYSTEMS (NASSAT)	ELIMCO AEROSPACE	SONOVISIÓN
AYTANA	FOSSA SYSTEMS	SPACE SUBMICRON ELECTRONICS 2SE
AERNNOVA	GMV INNOVATING SOLUTIONS, S.L.	TEST IN SPACE
AEROSERTEC	GRS	TRIGO GLOBAL Q
AERTEC SOLUTIONS	GRUPO EMERGY	TTI, INFORMATION AND COMMUNICATION TECHNOLOGIES

Source : Own elaboration

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